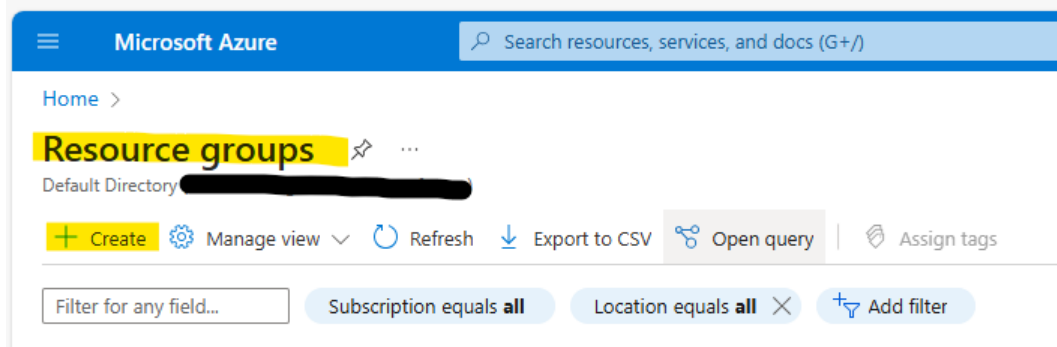


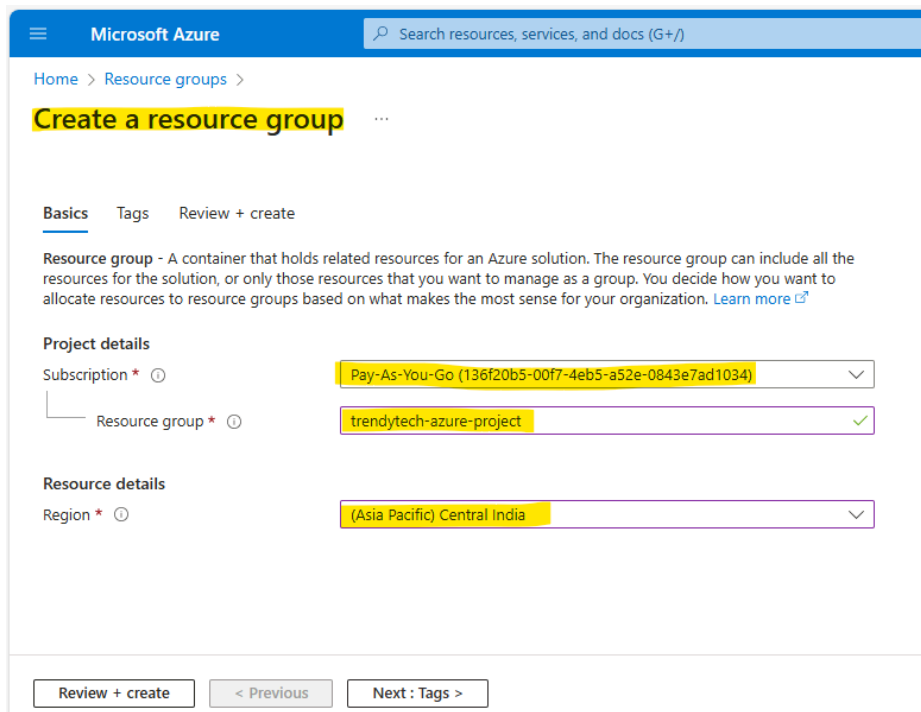
Part 1:

Before creating all the resources we will create the resource group in which we will create all the required resources.

Go to Azure Portal and log in with your Azure account. In the left-hand menu, select "Resource groups". If you don't see it, use the search bar at the top of the page and search for "Resource groups". Click the "+ Create" button or "Add" at the top of the Resource groups page.



Select the subscription under which the resource group will be created. Enter a unique name for your resource group. Choose a location (region) where your resources will reside (e.g., East US, West Europe).



Click "Review + Create" and then "Create".

Azure Storage Account creation:

Search for "Storage accounts" and click "+ Create". Select subscription, resource group, region, and enter a unique storage account name as shown below.

Home > Storage accounts >

Create a storage account

manage your storage account together with other resources.

Subscription *

Resource group * [Create new](#)

Instance details

Storage account name *

Region * [Deploy to an Azure Extended Zone](#)

Primary service

Performance * ☒ Standard: Recommended for most scenarios (general-purpose v2 account)
☐ Premium: Recommended for scenarios that require low latency.

Redundancy *

[Previous](#) [Next](#) [Review + create](#)

Enable ADLS Gen2: Go to the Advanced tab and enable Hierarchical namespace.

Home > Storage accounts >

Create a storage account

Basics **Advanced** Networking Data protection Encryption Tags Review + create

Security

Configure security settings that impact your storage account.

Require secure transfer for REST API operations ☒

Allow enabling anonymous access on individual containers ☐

Enable storage account key access ☒

Default to Microsoft Entra authorization in the Azure portal ☐

Minimum TLS version

Permitted scope for copy operations (preview)

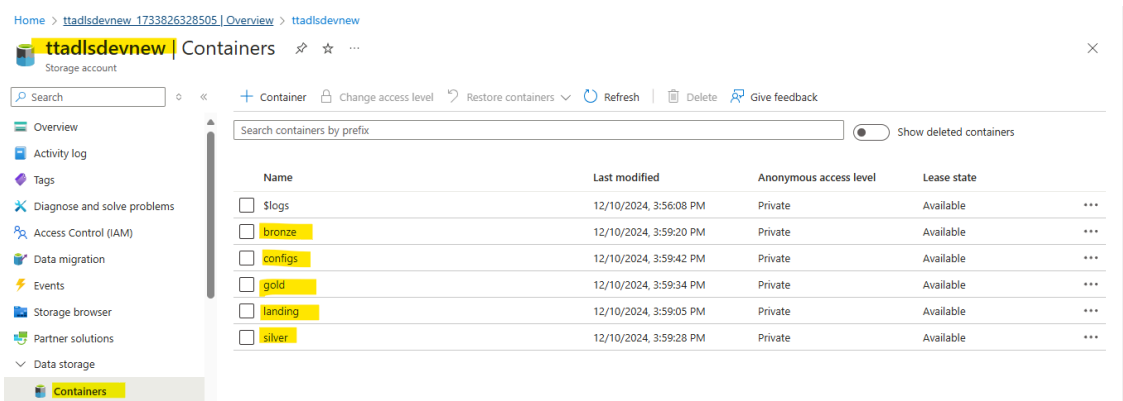
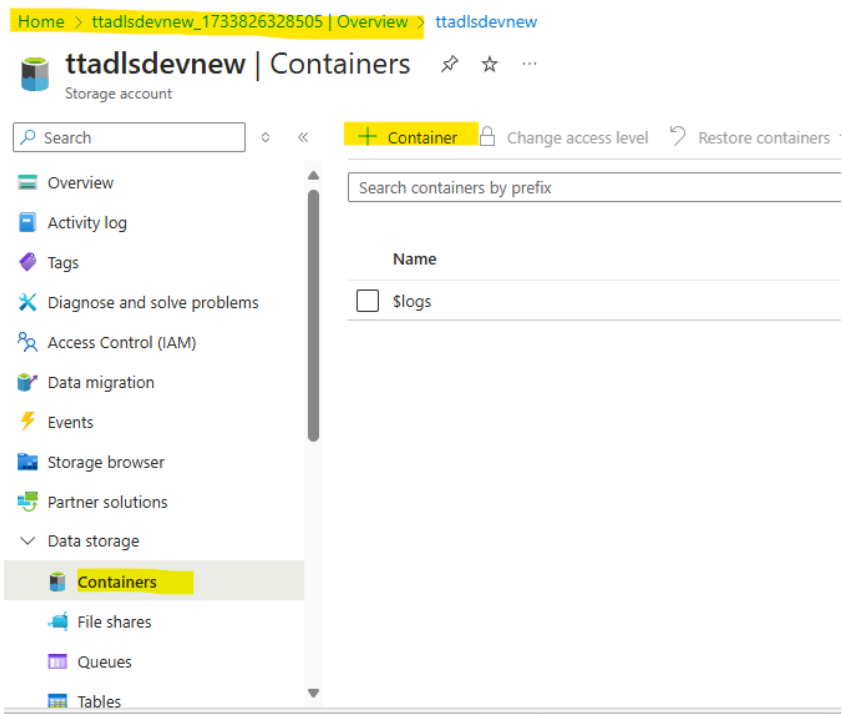
Hierarchical Namespace

Hierarchical namespace, complemented by Data Lake Storage Gen2 endpoint, enables file and directory semantics, accelerates big data analytics workloads, and enables access control lists (ACLs) [Learn more](#)

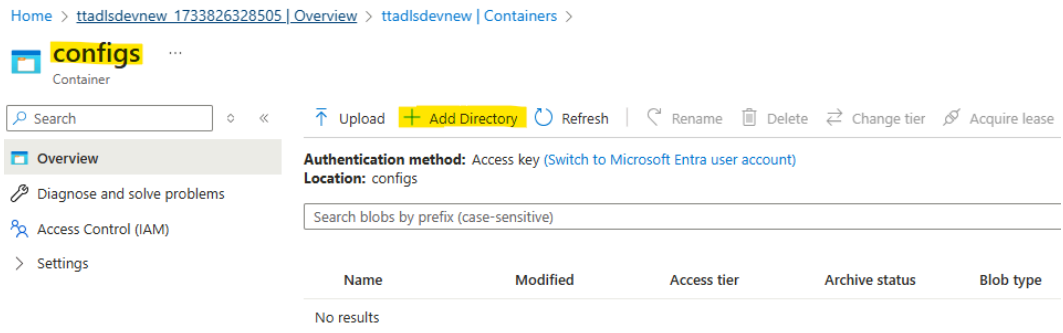
Enable hierarchical namespace ☒

Click Review + create, validate settings, and click Create.

Now create the containers “landing”, “bronze”, “silver”, “gold”, “configs” in this storage account as shown below.



Then in the configs container create the directory “emr” and then upload the file “load_config.csv” in it.



Home > ttadlsdevnew_1733826328505 | Overview > ttadlsdevnew | Containers >

configs

Container

Search

UploadAdd DirectoryRefreshRenameDeleteChange tierAcquire leaseBreak lease

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: configs / emr

Search blobs by prefix (case-sensitive)

Show

Name	Modified	Access tier	Archive status	Blob type	Size
[.]					
load_config.csv	12/10/2024, 4:04:45 ...	Hot (Inferred)		Block blob	743 B

Steps to create Azure SQL database:

We will create 2 azure SQL db - trendytech-hospital-a, trendytech-hospital-b

In the search bar, type "SQL Database" and select "SQL Database" from the results. Choose your Subscription and Resource Group. Enter a Database Name. Also create a SQL Server.

Home > SQL databases >

Create SQL Database

Microsoft

Want to try Azure SQL Database for free? Create a free serverless database with the first 100,000 vCore seconds, 32GB of data, and 32GB of backup storage free per month for the lifetime of the subscription. [Learn more](#)

Apply offer (Preview)

SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *

Pay-As-You-Go (136f20b5-00f7-4eb5-a52e-0843e7ad1034)

Resource group *

trendytech-azure-project

Create new

Database details

Enter required settings for this database, including picking a logical server and configuring the compute and storage resources

Database name *	trendytech-hospital-a
Server * ⓘ	Select a server Create new
Compute + storage * ⓘ	Please select a server first. Configure database

We will create the server as shown below. Choose a Compute + Storage tier (e.g., Basic, General Purpose).

[Home](#) > [SQL databases](#) > [Create SQL Database](#) >

Create SQL Database Server ...

Microsoft

Server details

Enter required settings for this server, including providing a name and location. This server will be created in the same subscription and resource group as your database.

Server name *	trendytech-sqlserver
Location *	(Asia Pacific) Central India

Authentication

 Azure Active Directory (Azure AD) is now Microsoft Entra ID. [Learn more](#)

Select your preferred authentication methods for accessing this server. Create a server admin login and password to access your server with SQL authentication, select only Microsoft Entra authentication [Learn more](#) using an existing Microsoft Entra user, group, or application as Microsoft Entra admin [Learn more](#), or select both SQL and Microsoft Entra authentication.

Authentication method

- ☐ Use Microsoft Entra-only authentication
- ☐ Use both SQL and Microsoft Entra authentication
- ☒ Use SQL authentication

Server admin login *	trendytechadmin
Password *	
Confirm password *	

OK

After this Go to Networking => In Network connectivity select “Public endpoint” option. Also set yes for the options “Allow Azure services and resources to access this server” and “Add current client IP address”

Basics **Networking** Security Additional settings Tags Review + create

Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'trendytech-sqlserver' and all databases it manages. [Learn more](#)

Firewall rules

The settings displayed below are read-only. They can be modified from the “Firewalls and virtual networks” blade for the selected server after database creation. [Learn more](#)

Allow Azure services and resources to access this server

No Yes

Add current client IP address *

No Yes

Private endpoints

Private endpoint connections are associated with a private IP address within a Virtual Network. The list below shows all the private endpoint connections for this server. Note that private endpoint connections are defined at the server level and they provide access to all databases in the server. [Learn more](#)

Note: Please note down this username and password for future reference.

[Home](#) > [SQL databases](#) >

Create SQL Database

Microsoft

Configure network access and connectivity for your server. The configuration selected below will apply to the selected server 'trendytech-server-new' and all databases it manages. [Learn more](#)

Network connectivity

Choose an option for configuring connectivity to your server via public endpoint or private endpoint. Choosing no access creates with defaults and you can configure connection method after server creation. [Learn more](#)

Connectivity method ⓘ

☐ No access

☒ Public endpoint

☐ Private endpoint

Firewall rules

Setting 'Allow Azure services and resources to access this server' to Yes allows communications from all resources inside the Azure boundary, that may or may not be part of your subscription. [Learn more](#)

Setting 'Add current client IP address' to Yes will add an entry for your client IP address to the server firewall.

Allow Azure services and resources to access this server *

No Yes

Add current client IP address *

No Yes

[Review + create](#) [< Previous](#) [Next : Security >](#)

Click Review + Create, validate the details, and then click Create as shown below.

Home > SQL databases >

Create SQL Database

Microsoft

Basics Networking Security Additional settings Tags Review + create

Create a SQL database with your preferred configurations. Complete the Basics tab then go to Review + Create to provision with smart defaults, or visit each tab to customize. [Learn more](#)

Free database offer applied! You got first 100,000 vCore seconds and 32GB of data & 32GB of backup storage free per month for lifetime of the subscription. [Learn more](#)

Remove offer

SQL Database Hyperscale: Low price, high scalability, and best feature set. [Learn more](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Review + create Next: Networking >

Cost summary

General Purpose (GP_S_Gen5_2)	
Cost per GB (in INR)	0.00
Max storage selected (in GB)	x 41.6
First 32 GB storage free	
First 100,000 vCore seconds free	
Overage billing¹	Disabled

Note: While creating database if you are not able to allow public access and add client ip address you can follow below steps:

After creating this database Go to Networking => For Public access (select option Selected networks and save this) =>

Home > Microsoft.SQLDatabase.newDatabaseExistingServer_906552b051fe4404 | Overview > trendytech-hospital-a (trendytech-sqlserver)

trendytech-sqlserver | Networking

SQL server

Search

- Overview
- Activity log
- Access control (IAM)
- Tags
- Quick start
- Diagnose and solve problems
- Settings
- Data management
- Security
- Networking**
- Microsoft Defender for Cloud

Feedback

Public access Private access Connectivity

Public network access

Public Endpoints allow access to this resource through the internet using a public IP address. An application requires proper authorization to access this resource. [Learn more](#)

Public network access

☐ Disable

☒ **Selected networks**

Connections from the IP addresses configured in the Firewall rules are allowed. [Learn more](#)

Virtual networks

Allow virtual networks to connect to your resource using service endpoints. [Learn more](#)

+ Add a virtual network rule

Also while using query editor if you face below error click on “Allowing IP for current ip address” as shown below



Similarly we will create another database trendytech-hospital-b(We will use the same server i.e trendytech-sqlserver that we have created while creating trendytech-hospital-a database). Thus we have created 2 databases as shown below.

Home >				
SQL databases  ...				
Default Directory (aratishatti15gmail.onmicrosoft.com)				
+ Create ⌚ Reservations ⚙️ Manage view 🔄 Refresh ⬇️ Export to CSV 🔗 Open query 🏷️ Assign tags 🗑️ Delete				
Filter for any field... Subscription equals all Resource group equals all ✕ Location equals all ✕ Add filter				
Showing 1 to 2 of 2 records. No grouping				
☐ Name ↑↓	Server ↑↓	Replica type ↑↓	Pricing tier ↑↓	Location ↑↓
☐ trendytech-hospital-a (trendytech-sqlserver/trendytech-hospital-a)	trendytech-...	--	General Purpose: S...	Central India
☐ trendytech-hospital-b (trendytech-sqlserver/trendytech-hospital-b)	trendytech-...	--	Free General Purpo...	Central India

Then we will create the tables in these databases and for creating tables in the database use below scripts which are present on github account:

For trendytech-hospital-a =>

Trendytech_hospital_A_table_creation_commands

For trendytech-hospital-b =>

Trendytech_hospital_B_table_creation_commands

Steps to create ADF:

In the search bar, type "Data Factory" and select "Data Factory" from the results. Click the "Create" button on the Data Factory page. Provide a globally unique name for your Data Factory instance. Choose V2 (Data Factory Version 2) for the latest features.

Home > Data factories > Create Data Factory ...

Basics Git configuration Networking Advanced Tags Review + create

One-click to create data factory with sample pipeline and datasets. [Try it](#)

Project details

Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription * ⓘ Pay-As-You-Go (136f20b5-00f7-4eb5-a52e-0843e7ad1034) ▼

Resource group * ⓘ trendytech-azure-project ▼
[Create new](#)

Instance details

Name * ⓘ TT-health-projectdev ✓

Region * ⓘ Central India ▼

Version * ⓘ V2 ▼

Previous Next Review + create

Click "Review + Create" to validate the details. If validation passes, click "Create" to deploy the Data Factory.

Steps to create ADF pipeline:

Linked Services creation:

In the ADF interface, go to the Manage section on the left-hand panel. Under the Connections section, select Linked Services. Click on New to create a new Linked Service.

1. Azure SQL DB

Note down the server name for that sql database that we have created.

Home >

 **trendytech-hospital-b (trendytech-sqlserver/trendytech-hospital-b)**   ...

SQL database

Search       

Overview

- Activity log
- Tags
- Diagnose and solve problems
- Query editor (preview)
- Mirror database in Fabric (preview)
- Settings
- Data management
- Integrations

Mirror databases in Microsoft Fabric Easily replicate your existing databases in Fabric, and help your team achieve streamlined ETL and operational analytics goals. [Learn more](#)

Essentials

Resource group (...)	: trendytech-azure-project	Server name	: trendytech-sqlserver.database.windows.net
Status	: Paused	Connection strings	: Show database connection strings
Location	: Central India	Pricing tier	: Free - General Purpose - Serverless: Gen5, 2 vC
Subscription (move)	: Pay-As-You-Go	Overage billing	: Disabled
Subscription ID	: 136f20b5-00f7-4eb5-a52e-0843e7ad1034	Free monthly vCore ...	: 99,424 vCore seconds remaining
Tags (edit)	: Add tags	Earliest restore point	: 2024-12-10 17:18 UTC

In fully qualified domain names, mention the server name, mention the username and password for sql server and define the parameter db_name and using this parameter we will pass the database name as shown below.

New linked service

 Azure SQL Database [Learn more](#) 

Name *

hosa_sql_ls

Description

Connect via integration runtime * 

 AutoResolveIntegrationRuntime

Version

☒ Recommended ☐ Legacy

 Import from connection string

Account selection method 

☐ From Azure subscription ☒ Enter manually

Fully qualified domain name *

trendytech-sqlserver.database.windows.net

Database name

@linkedService().db_name

Authentication type *

SQL authentication

User name *

trendytechadmin

Password

Azure Key Vault

Password *

.....

Always encrypted ⓘ

☐

Encrypt ⓘ

Mandatory

Trust server certificate ⓘ

☐

Host name in certificate

Additional connection properties

+ New

Annotations

+ New

Parameters

+ New | Delete

☐	Name	Type	Default value	
☐	db_name	String	Value	

> Advanced ⓘ

Create

Back

Test connection

Cancel

And click on create to create the linked service.

2. ADLS GEN2

Select Azure Data Lake Storage1 as the data store. Provide the following details- Name of your Blob Storage account, Authentication. Then click Test Connection to verify and save the Linked Service.

To get the url for Azure Data Lake Storage go to Adls gen2 storage that we have created => Setting => Endpoints => and copy the URL as shown below

Home > ttadlsdevnew

ttadlsdevnew | Endpoints

Storage account

Search Refresh Give feedback

- Data storage
- Security + networking
- Data management
- Settings**
 - Configuration
 - Resource sharing (CORS)
 - SFTP
 - Advisor recommendations
 - Endpoints**
 - Locks

Resource ID	/subscriptions/136f20b5-00f7-4eb5-a52e-0843e7ad10
Queue service	https://ttadlsdevnew.queue.core.windows.net/
Table service	
Resource ID	/subscriptions/136f20b5-00f7-4eb5-a52e-0843e7ad10
Table service	https://ttadlsdevnew.table.core.windows.net/
Data Lake Storage	
Resource ID	/subscriptions/136f20b5-00f7-4eb5-a52e-0843e7ad10
Data Lake Storage	https://ttadlsdevnew.dfs.core.windows.net/

Also copy the access key. Using these details create the linked service as shown below.

New linked service

Azure Data Lake Storage Gen2 [Learn more](#)

Name *
AzureDataLakeStorage1

Description

Connect via integration runtime * ⓘ
☒ AutoResolveIntegrationRuntime

Authentication type
Account key

Account selection method ⓘ
☐ From Azure subscription ☒ Enter manually

URL *
https://ttadlsdevnew.dfs.core.windows.net/

☒ Storage account key ☐ Azure Key Vault

Storage account key *
.....

Create Back Test connection Cancel

3. Delta table - Audit_logs

Create the databricks workspace test and then upload the code notebook “Audit_table_DDL” and start your databricks cluster and create the schema audit table in it using below commands. (Notebook name - audit_table_ddl)

create schema if not exists audit;

```
CREATE TABLE IF NOT EXISTS audit.load_logs (  
  id BIGINT GENERATED ALWAYS AS IDENTITY,  
  data_source STRING,  
  tablename STRING,  
  numberofrowscopied INT,  
  watermarkcolumnname STRING,  
  loaddate TIMESTAMP  
);
```

Home > Azure Databricks >

Create an Azure Databricks workspace ...

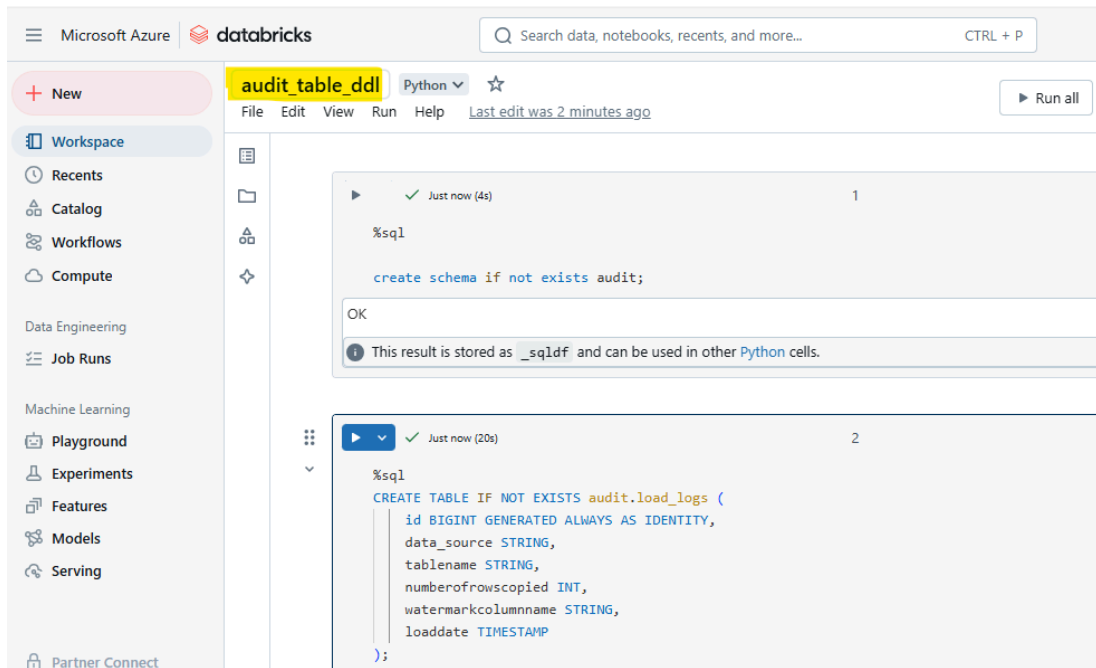
Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.

Subscription *	Pay-As-You-Go (136f20b5-00f7-4eb5-a52e-0843e7ad1034) ▼
Resource group *	trendytech-azure-project ▼ Create new
Instance Details	
Workspace name *	test ✓
Region *	Central India ▼
Pricing Tier *	Standard (Apache Spark, Secure with Microsoft Entra ID) ▼
Managed Resource Group name	Enter name for managed resource group

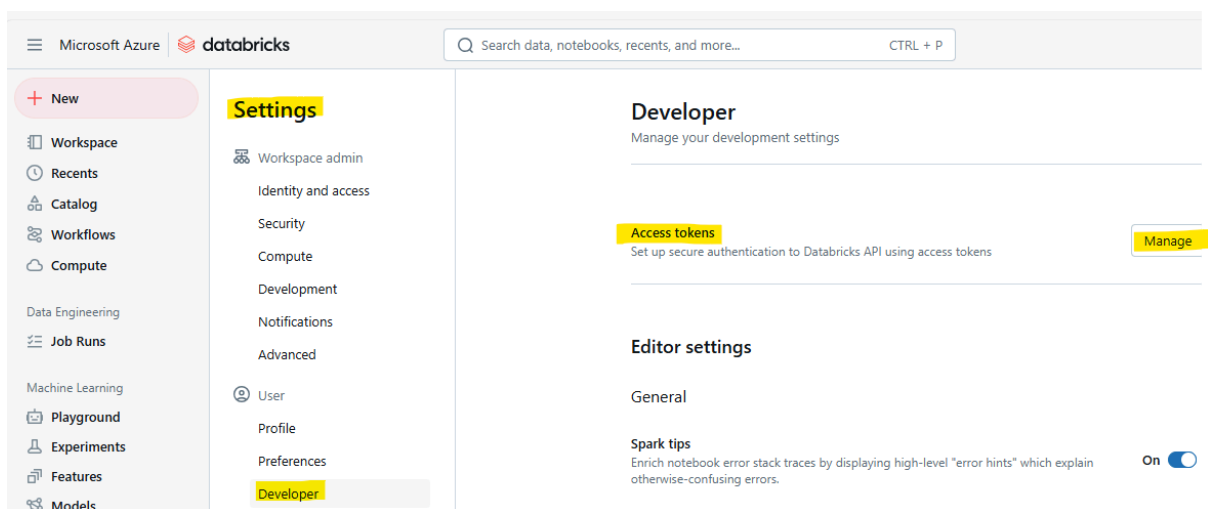
[Review + create](#)

[< Previous](#)

[Next : Networking >](#)

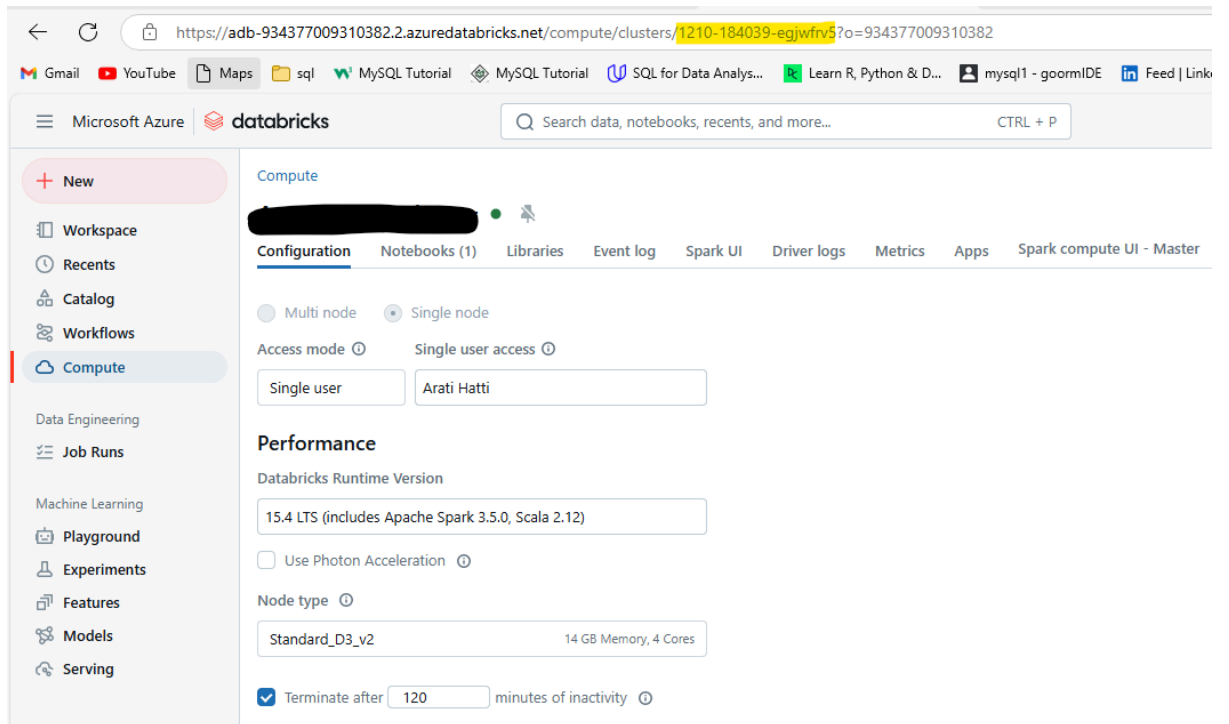


Note: To get access token Click your profile icon (top-right corner of the workspace) => Select Settings => Developer (In user setting) => Generate Access Token as shown below



Then generate the token and copy for Future use.

While creating linked service in source mention “AzureDatabricksDeltaLake”. Then in the domain mention the URL of databricks workspace. And mention the cluster id for the cluster that we have created. To get Databricks overview page to get Workspace URL and to get cluster id go to compute => Select the cluster => copy the cluster id as shown below



Refer below screenshot for more details.

New linked service
Azure Databricks Delta Lake [Learn more](#)

Name *
AzureDatabricksDeltaLake1_Is

Description

Connect via integration runtime * ⓘ
✓ AutoResolveIntegrationRuntime

Authentication method *
Access token

Account selection method
☐ From Azure subscription ☒ Enter manually

Domain *
https://adb-934377009310382.2.azuredatabricks.net

Existing cluster ID *
1210-184039-eqjwfrv5

✓ Connection successful
[Test connection](#) [Cancel](#)

[Create](#) [Back](#)

Dataset creation:

In the ADF interface, click on the Author section (left-hand panel). Expand the Datasets option. Click on the “...” next to Datasets in order to create the dataset.

1. Azure SQL DB

We will select the linked service that we have created for the SQL database. To create the datasets for the tables in a parameterized way in the sql database , we will create the parameter db_name, schema_name, table_name.

Set properties

Name

generic_sql_ds

Linked service *

hosa_sql_ls

Table name

Select...

☐ Enter manually

Import schema

☐ From connection/store

☒ None

OK

Back

Cancel

Then we will create parameters db_name, schema_name, table_name as shown below.

generic_sql_ds



Azure SQL Database
generic_sql_ds

Connection
Schema
Parameters

+ New
Delete

☐	Name	Type	Default value	
☐	db_name	String	Value	
☐	schema_name	String	Value	
☐	table_name	String	Value	

And we will pass dynamic value for table name and schema name as shown below

generic_sql_ds



Azure SQL Database
generic_sql_ds

Connection
Schema
Parameters

Linked service *
hosa_sql_ls

Test connection
Edit
+ New
Learn more

Linked service properties ⓘ

Name	Value	Type
db_name	@dataset().db_name	string

Table

@dataset().schema_name

@dataset().table_name

☒ Enter manually

Preview data

2. Dataset for Flatfile in ADLS GEN2

Select source as ADLS gen2 and file format as delimited text. Also in order to make it generic we will create the parameter file_name, file_path and container.

Set properties

Name
generic_adls_flat_file_ds

Linked service *
AzureDataLakeStorage1

File path
File system / Directory / File name

First row as header ☒

Import schema
☐ From connection/store ☐ From sample file ☒ None

OK Back Cancel

generic_sql_ds x generic_adls_flat_fil..

DelimitedText
generic_adls_flat_file_ds

Connection Schema Parameters

+ New Delete

☐	Name	Type	Default value	
☐	file_name	String	Value	
☐	file_path	String	Value	
☐	container	String	Value	

generic_sq_ds

generic_adls_flat_fil...



DelimitedText
generic_adls_flat_file_ds

Connection

Schema

Parameters

Linked service *

AzureDataLakeStorage1

Test connection

Edit

New

Learn more

File path

@dataset().container

/

@dataset().file_path

/

@dataset().file_name

Compression type

No compression

Column delimiter

Comma (,)

Row delimiter

Default (\r\n, or \n)

Encoding

Default(UTF-8)

Quote character

Double quote (")

Escape character

Backslash (\)

First row as header

☒

Now publish the changes.

3. Dataset for Parquet file in ADLS GEN2

In order to store data in ADLS gen2 in parquet format we will need the dataset.

While creating this dataset we will select source as ADLS gen2, fileformat as parquet and we will create parameters file_name, file_path and container.

Set properties

Name

generic_adls_parquet_ds

Linked service *

AzureDataLakeStorage1

File path

File system

/

Directory

/

File name

Import schema

☐ From connection/store
 ☐ From sample file
 ☒ None

OK

Back

Cancel

generic_sql_ds
generic_adls_flat_file...
generic_adls_parquet...


Parquet
generic_adls_parquet_ds

Connection
Schema
Parameters

Linked service *
AzureDataLakeStorage1
Test connection
Edit
New
Learn more

File path
@dataset().container
/
@dataset().file_path
/
@dataset().file_name

Compression type
snappy

4. Databricks Delta Lake for Delta lake

We will select the source as Azure Databricks Delta Lake. For this we will create the parameter schema_name and table_name.

generic_sql_ds
generic_adls_flat_file...
generic_adls_parquet...
AzureDatabricksDelt...


Azure Databricks Delta Lake
AzureDatabricksDeltaLakeDataset1

Connection
Schema
Parameters

Linked service *
AzureDatabricksDeltaLake1_Is
Test connection
Edit
New
Learn more

Database
@dataset().schema_name

Table
@dataset().table_name
Preview data

Once all the dataset and linked service are created, publish all in order to save them.

Creation of Pipelines:

Background activity : Creation of pipeline to copy data into sql tables (pl_to_insert_data_to_sql_table_preprocessing).

Before proceeding with the main pipeline, we will create a simple pipeline in Azure Data Factory (ADF) to copy data from ADLS Gen2 storage into tables in an SQL database. This serves as a prerequisite to ensure that the SQL tables contain the data needed for the main pipeline.

Note: We will create a new container (raw-data-for-sql-database) in the given ADLS Gen2 storage (adlsdevnew) and upload our CSV files, which will serve as the source for the pipeline, along with a lookup file. Additionally, we will create a dataset for the lookup file to use in this pipeline. Using the copy activity in ADF, we will transfer the data into the following tables: Departments, Providers, Encounters, Patients, and Transactions, located in the SQL databases trendytech-hospital-a and trendytech-hospital-b.

Source: ADLS gen2 -adlsdevnew

We will create a new container (raw-data-for-sql-database) in the given ADLS Gen2 storage (adlsdevnew) and upload our CSV files, along with a lookup file.

Home > ttadlsdevnew

ttadlsdevnew Containers

Storage account

Search

Overview
Activity log
Tags
Diagnose and solve problems
Access Control (IAM)
Data migration
Events
Storage browser
Partner solutions
Data storage

Container Change access level Restore containers Refresh Delete Give feedback

Search containers by prefix

Name	Last modified
☐ \$logs	12/10/2024, 3:56:08 PM
☐ bronze	12/10/2024, 3:59:20 PM
☐ configs	12/10/2024, 3:59:42 PM
☐ gold	12/10/2024, 3:59:34 PM
☐ landing	12/10/2024, 3:59:05 PM
☒ raw-data-for-sql-database	12/11/2024, 4:40:31 PM
☐ silver	12/10/2024, 3:59:28 PM

Folder: HospitalA, HospitalB for datafiles, Lookup for lookup file

Home > ttadlsdevnew | Containers >

raw-data-for-sql-database
Container

Search

UploadAdd DirectoryRefreshRenameDeleteChange ti

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: raw-data-for-sql-database

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier
☐ HospitalA	12/11/2024, 2:56:39 ...	
☐ HospitalB	12/11/2024, 2:56:48 ...	
☐ Lookup	12/13/2024, 8:24:50 ...	

Sink: SQL DB - trendytech-hospital-a, trendytech-hospital-b:

Note: We have already created these databases so no need to create again.

Home >

SQL databases

Default Directory (aratishatti15gmail.onmicrosoft.com)

CreateReservationsManage viewRefreshExport to CSVOpen queryAssign tagsDelete

Filter for any field...Subscription equals allResource group equals allLocation equals allAdd filter

Showing 1 to 2 of 2 records.

☐ Name ↑↓	Server ↑↓	Replica type ↑↓
☐ trendytech-hospital-a (trendytech-server-new/trendytech-hospital-a)	trendytech-...	--
☐ trendytech-hospital-b (trendytech-server-new/trendytech-hospital-b)	trendytech-...	--

Pipeline creation Steps:

1. Creation of Linked Services:
- => For ADLS gen2 storage(source):

We will use the same linked service “AzureDataLakeStorage1” that we have created earlier for ADLS Gen2 storage.

=> For SQL DB(Sink):

We will use the same linked service “hosa_sql_ls” that we have created earlier for the database.

2. Creation of Datasets:

=> For source:

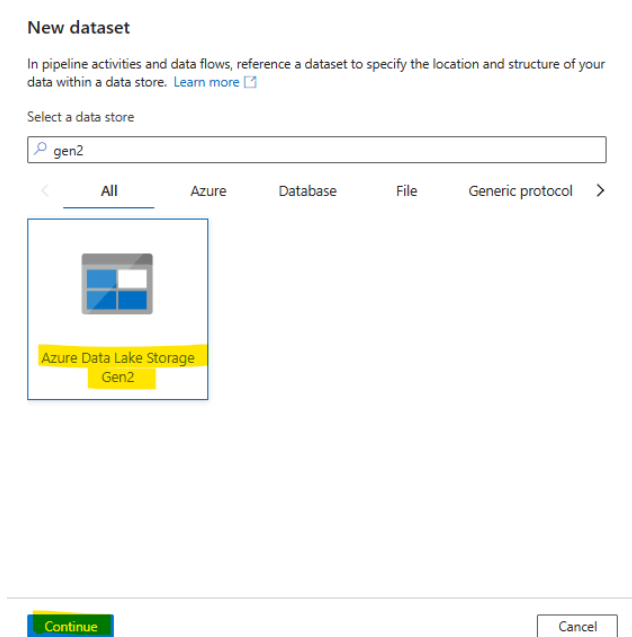
We will use the same generic dataset “generic_adls_flat_file_ds” that we have created earlier.

=> For sink:

We will use the same generic dataset “generic_sql_ds” that we have created earlier.

=>For Lookup we will create a new dataset as shown below.

Select source as ADLS Gen2 storage, then in file format select json as our lookup file is a json file as shown below



Select format

Choose the format type of your data

 Avro	 Binary	 DelimitedText
 Excel	 Iceberg	 JSON
		 XML

Continue Back Cancel

Set properties

Name

ls_for_lookup_file_adls_to_sql

Linked service *

AzureDataLakeStorage1

File path

raw-data-for-sql-datab...

/ Lookup

/ Lookup_file_table_map...

Folder icon | Down arrow

Import schema

☐ From connection/store

☐ From sample file

☒ None

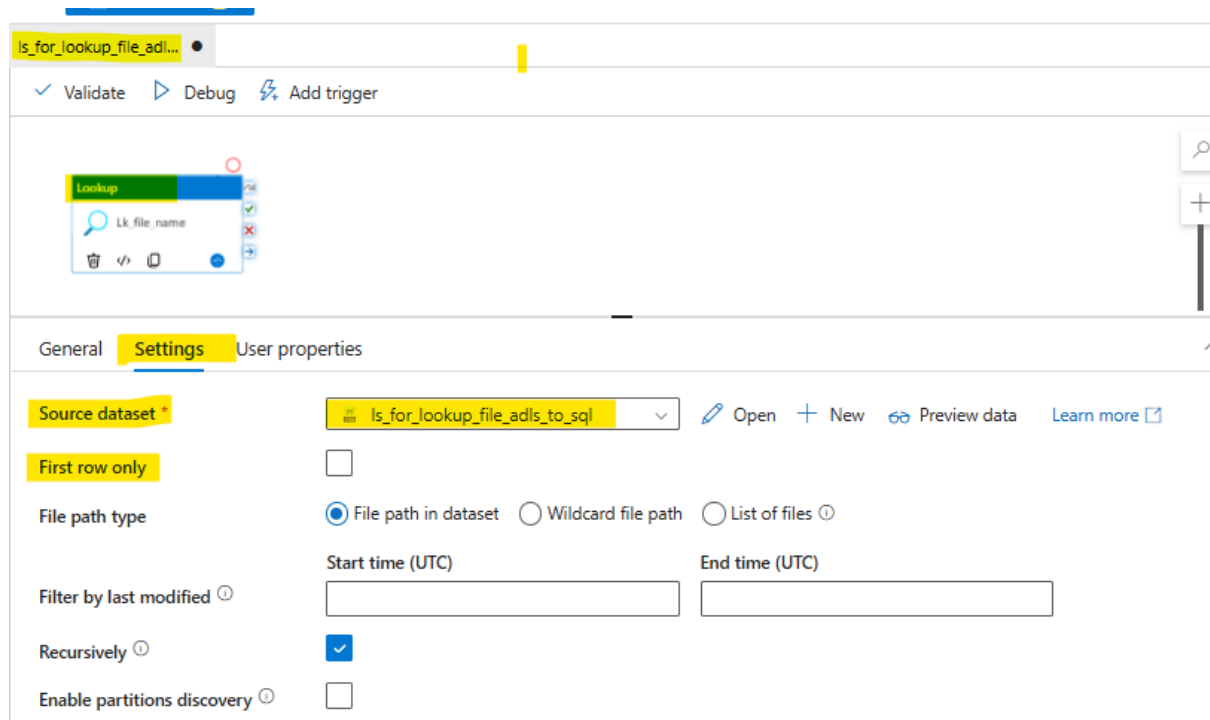
OK Back

Cancel

Steps to Configure the Pipeline:

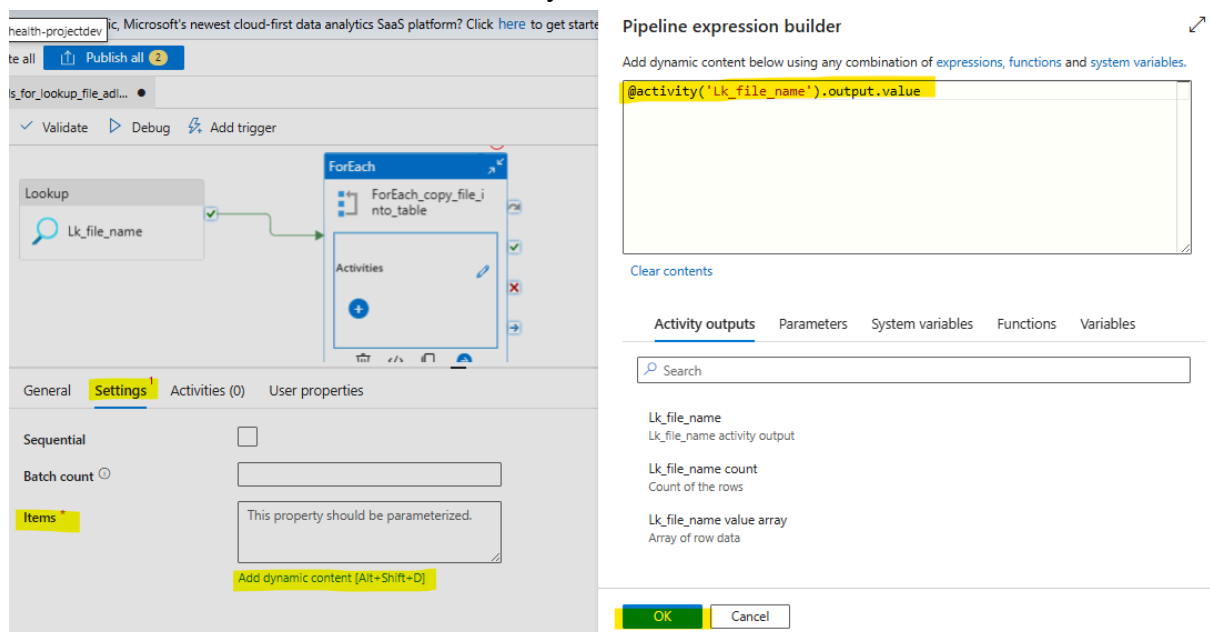
Add a Lookup Activity:

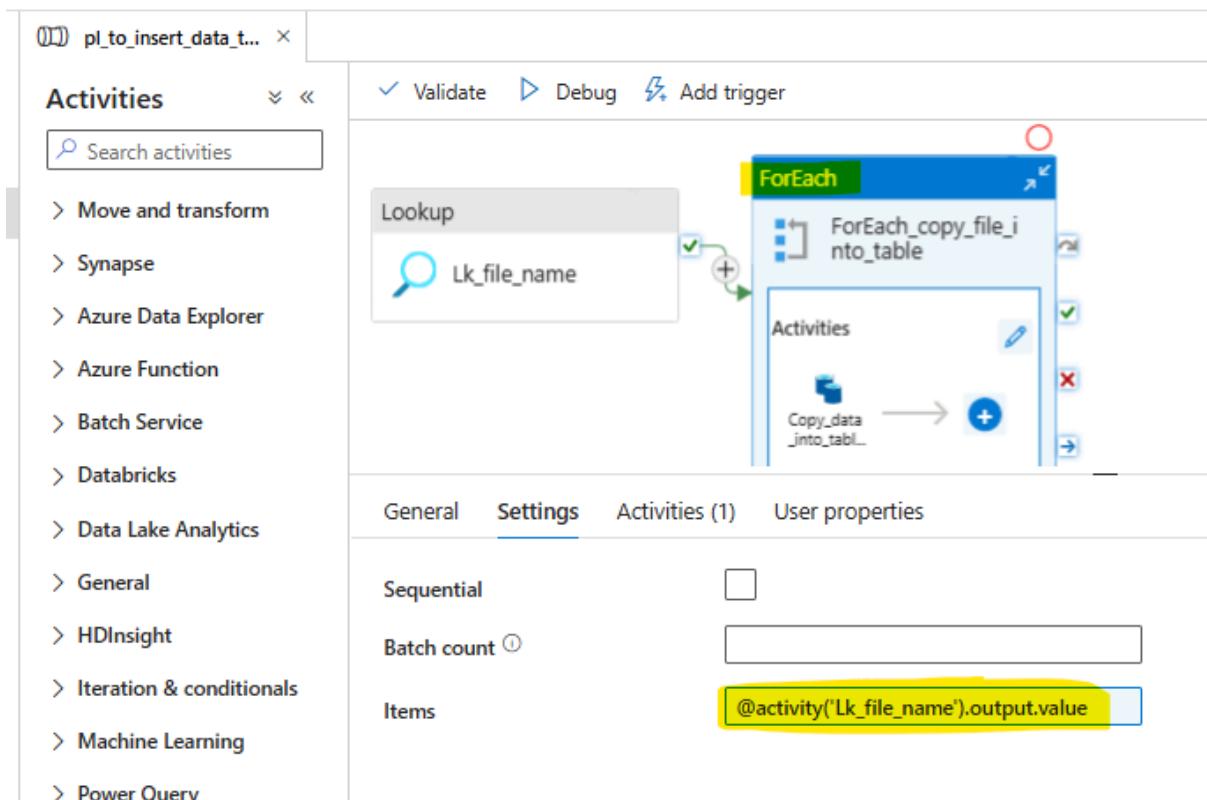
Drag a Lookup activity to the canvas. Point it to the mapping CSV dataset. Set First row only to false to read all rows. Refer below screenshot for more clarity.



Add a ForEach Activity:

Drag a ForEach activity and connect it to the Lookup activity. Set its Items property to `@activity('Lk_file_name').output.value`. Refer below screenshot for more clarity.

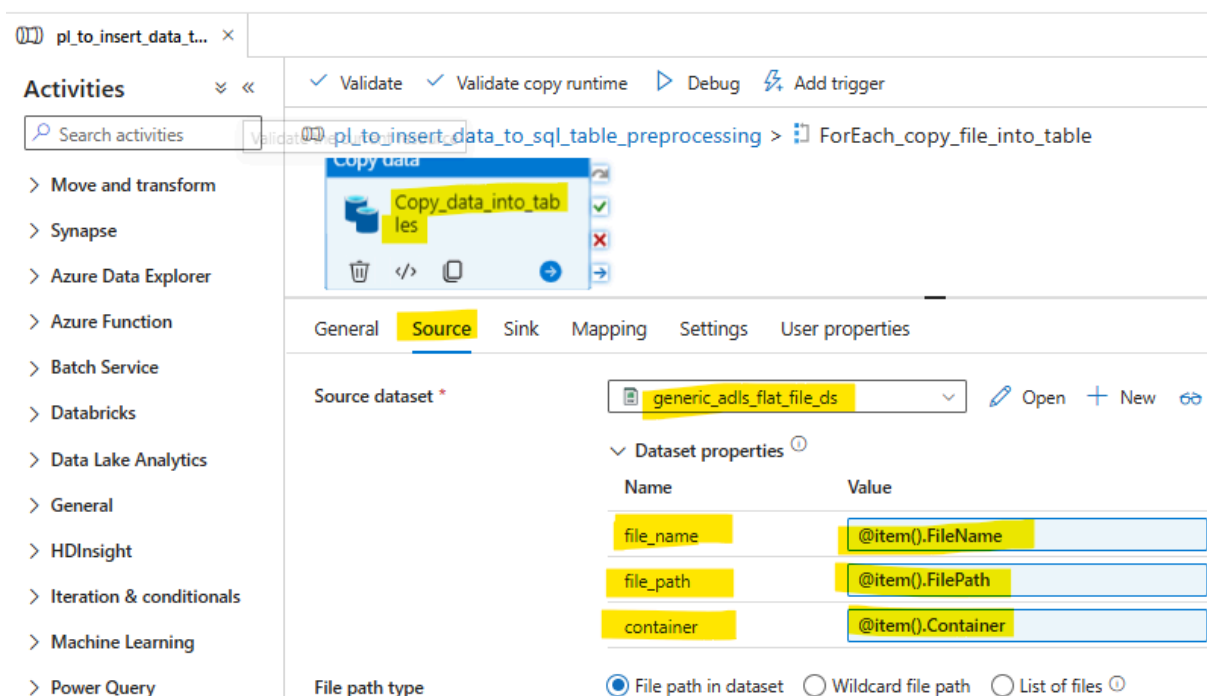




Configure the ForEach Activity:

Inside the ForEach activity, add a Copy Data activity.

Source: Use the source dataset.



Sink: Use the destination dataset.

The screenshot shows the configuration of a Sink dataset in Azure Data Factory. The 'Sink' tab is selected, and the 'Sink dataset' is set to 'generic_sql_ds'. The 'Dataset properties' table is as follows:

Name	Value	Type
db_name	@item().DatabaseName	string
schema_name	dbo	string
table_name	@item().TableName	string

This pipeline will copy the data from the file into the tables in the sql database. On successfully running the pipeline we will get below output.

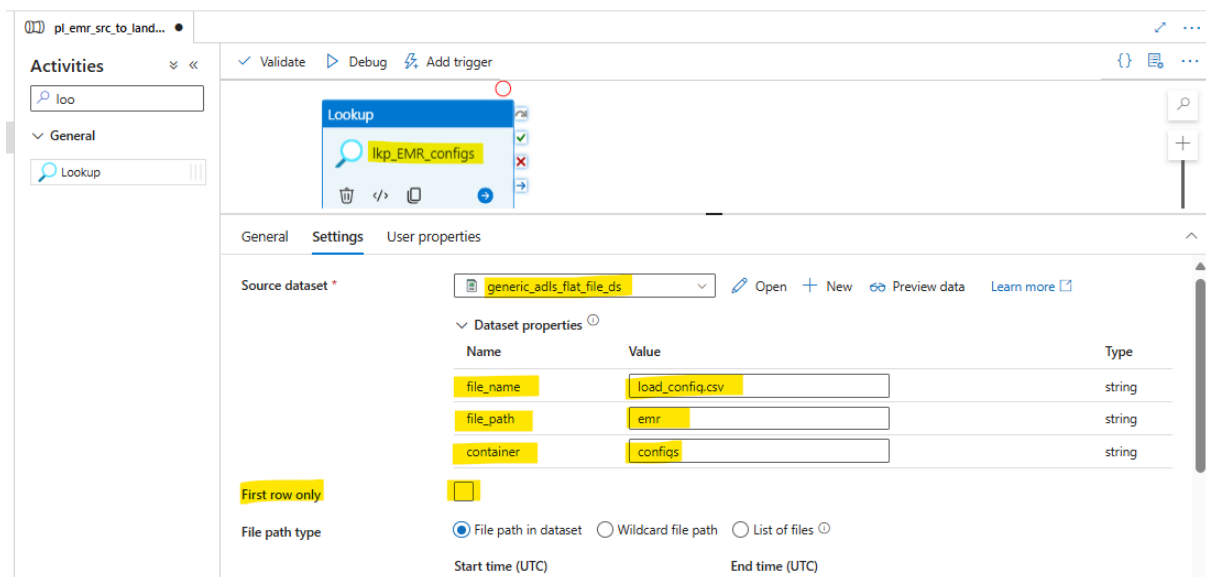
The screenshot shows the 'Output' tab of a pipeline run. The pipeline status is 'Succeeded'. The table below lists the activities and their results:

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime
Copy_data_into_tables	✓ Succeeded	Copy data	12/13/2024, 12:57:15 P	14s	AutoResolveIntegra
Copy_data_into_tables	✓ Succeeded	Copy data	12/13/2024, 12:57:15 P	14s	AutoResolveIntegra
Copy_data_into_tables	✓ Succeeded	Copy data	12/13/2024, 12:57:15 P	14s	AutoResolveIntegra
Copy_data_into_tables	✓ Succeeded	Copy data	12/13/2024, 12:57:15 P	14s	AutoResolveIntegra
Copy_data_into_tables	✓ Succeeded	Copy data	12/13/2024, 12:57:15 P	14s	AutoResolveIntegra
Copy_data_into_tables	✓ Succeeded	Copy data	12/13/2024, 12:57:15 P	13s	AutoResolveIntegra
Copy_data_into_tables	✓ Succeeded	Copy data	12/13/2024, 12:57:15 P	15s	AutoResolveIntegra
Copy_data_into_tables	✓ Succeeded	Copy data	12/13/2024, 12:57:15 P	14s	AutoResolveIntegra
Copy_data_into_tables	✓ Succeeded	Copy data	12/13/2024, 12:57:15 P	14s	AutoResolveIntegra

Pipeline to copy data from Azure Sql db to Landing Folder in ADLS Gen2

1. To read the config file we will use Lookup activity.

In this for source dataset will be for configs file and we will pass the parameter values as shown below.

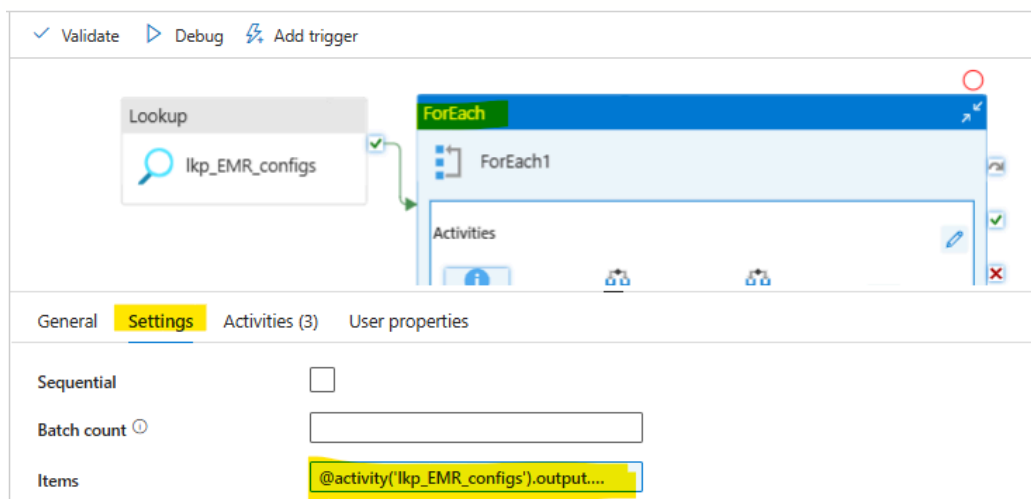


Then additionally we can preview the data.

2. In order to iterate through each row of configuration data we will use ForEach Activity.

Processing Logic Within ForEach Activity:

`@activity('lkp_EMR_configs').output.value`



a. We will use get metadata activity in order to check whether file exists in Bronze container:

To file name we will use below logic - `@split(item().tablename, '.')[1]`

file_path is present in lookup file as targetpath

And container name we will explicitly mention as bronze as shown below

The screenshot shows an Azure Data Factory workflow. A 'Lookup' activity named 'lkp_EMR_configs' is connected to a 'ForEach' loop named 'ForEach1'. Inside the 'ForEach' loop, there is an 'Activities' section containing a 'fileExists' activity, followed by an 'If Condition1' activity, and then a plus sign indicating more activities can be added. Below the workflow, the 'Settings' tab for the 'fileExists' activity is displayed. It shows the 'Dataset' as 'generic_adls_parquet_ds'. Under 'Dataset properties', there is a table with the following data:

Name	Value	Type
file_name	<code>@split(item().tablename, '.')[1]</code>	string
file_path	<code>@item().targetpath</code>	string
container	bronze	string

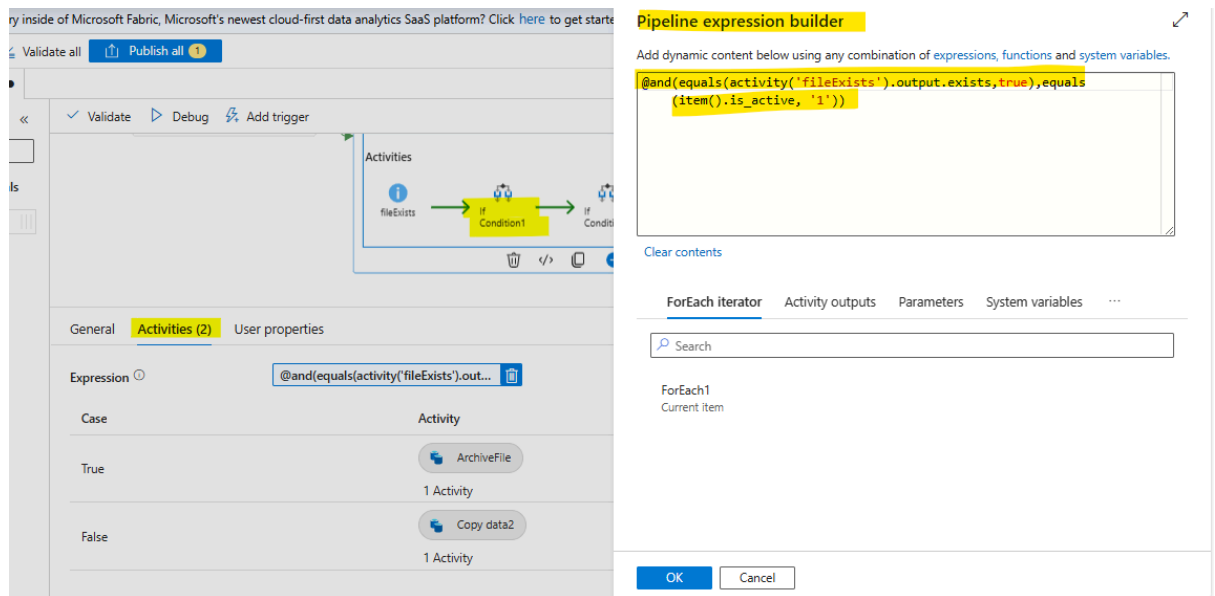
This will check if the file exists in the Bronze container. Based on the file's presence or absence, we will use an If Condition activity to determine the subsequent processing steps.

b. Use an If Condition activity based on the file's existence.

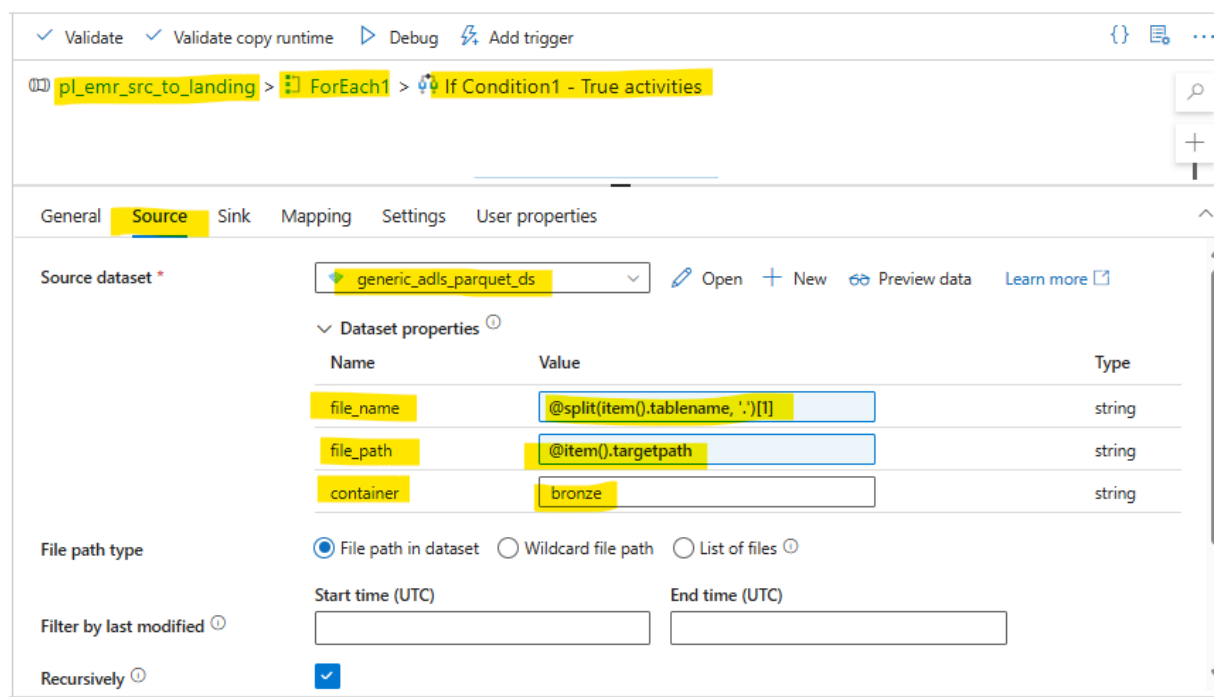
Condition 1: File Exists (True) => Move the file to the Archive folder.

condition:

`@and(equals(activity('fileExists').output.exists,true),equals(item().is_active, '1'))`



Source: Container: Bronze, Path: hosa, File: encounters



Target: Container: Bronze,

File_path -Path: hosa/archive/<year>/<month>/<day> =>
`@concat(item().targetpath, '/archive/',
formatDateTime(utcNow(), 'yyyy'), '/',
formatDateTime(utcNow(), '%M'), '/',
formatDateTime(utcNow(), '%d'))`

File_name - @split(item().tablename, '.')[1]

The screenshot shows the Microsoft Fabric pipeline editor. The main canvas displays a pipeline with a 'fileExists' activity and an 'ArchiveFile' activity. The 'Sink dataset' properties are visible, showing the 'file_name' property set to '@split(item().tablename, '.')[1]' and the 'file_path' property set to '@concat(item().targetpath, '/a'. The 'Pipeline expression builder' is open on the right, showing the expression '@concat(item().targetpath, '/archive/', formatDateTime(utcNow(), 'yyyy'), '/', formatDateTime(utcNow(), '%M'), '/', formatDateTime(utcNow(), '%d'))'.

c. Determine if it's a full load or incremental load using If condition.

@equals(items().loadtype, 'Full')

The screenshot shows the Microsoft Fabric pipeline editor. The main canvas displays a pipeline with a 'fileExists' activity, an 'If Condition1' activity, and a 'Condi' activity. The 'If Condition1' activity is highlighted, and its 'Expression' property is set to '@equals(items().loadtype, 'Full')'. The 'Pipeline expression builder' is open on the right, showing the expression '@equals(items().loadtype, 'Full')'. A warning message states: "'items' is not a recognized function".

If "If condition" holds true => Full Load => Copy all data from the database table. => Enter Log details in the audit table:

Folder and File Structure

Bronze Container:

Source Path: bronze/hosa

Target Path for Data Loads: bronze/<target-path>

✓ Validate ✓ Validate copy runtime ▶ Debug ⚡ Add trigger

pl_emr_src_to_landing > ForEach1 > If Condition2 - True activities

Copy data

Full_load_cp

General Source Sink Mapping Settings User properties

Source dataset * generic_sql_ds Open + New Preview

Dataset properties

Name	Value
db_name	@item().database
schema_name	@split(item().tablename, '.')[0]
table_name	@split(item().tablename, '.')[1]

pl_emr_src_to_landing > ForEach1 > If Condition2 - True activities

Copy data

Full_load_cp

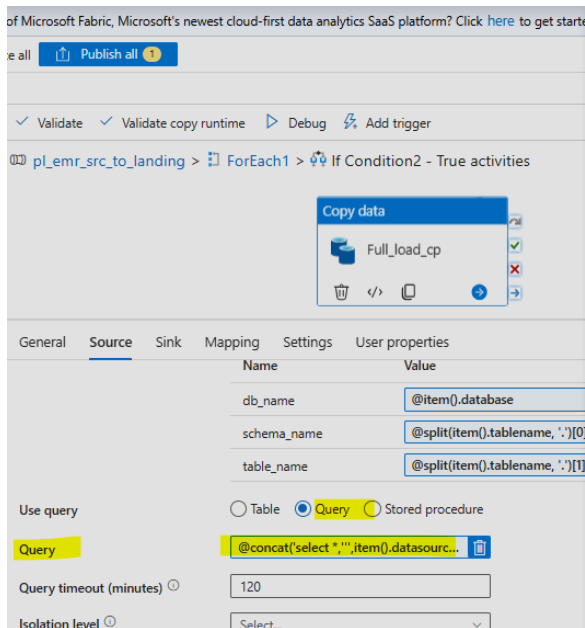
General Source Sink Mapping Settings User properties

Sink dataset * generic_adls_parquet_ds Open + New Learn more

Dataset properties

Name	Value
file_name	@split(item().tablename, '.')[1]
file_path	@item().targetpath
container	bronze

Query: @concat('select *,'',item().datasource, '' as datasource from ',item().tablename)



Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables.

```
@concat('select *,',item().datasource,' as datasource from ', item().tablename)
```

Clear contents

Activity outputs Parameters System variables Functions Variables

Search

ArchiveFile
ArchiveFile activity output

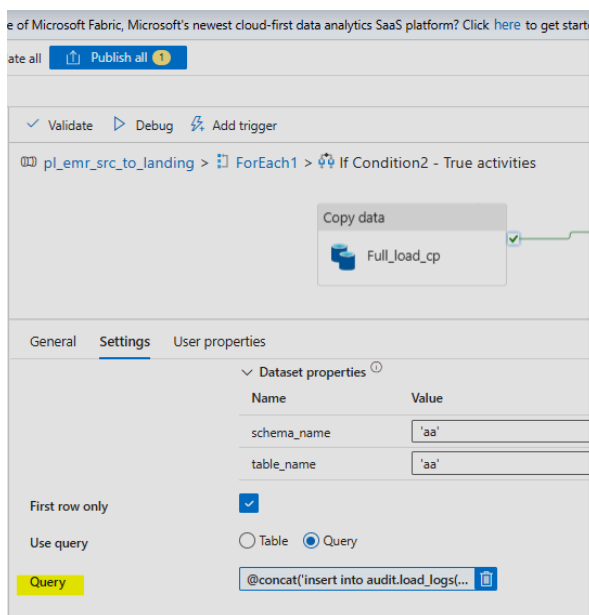
fileExists
fileExists activity output

fileExists childItems
List of subfolders and files in the given folder

OK Cancel

Enter Log details in the audit table:

Query: @concat('insert into audit.load_logs(data_source,tablename,numberofrowscopied,watermarkcolumnname,loaddate) values ('',item().datasource,'',item().tablename,'',activity('Full_Load_CP').output.rowscopied,'',item().watermark,'',utcNow(),''))')



Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables.

```
@concat('insert into audit.load_logs(data_source,tablename, numberofrowscopied,watermarkcolumnname,loaddate) values ('', item().datasource,'', item().tablename,'',activity ('Full_Load_CP').output.rowscopied,'',item().watermark, '',utcNow(),''))')
```

Clear contents

Activity outputs Parameters System variables Functions Variables

Search

ArchiveFile
ArchiveFile activity output

Full_load_cp
Full_load_cp activity output

fileExists
fileExists activity output

OK Cancel

If condition is false => Incremental Load

(Fetch incremental data using the last fetched date) using Lookup=>
Incremental load using copy activity => Enter log details in the audit table:

Lookup:

The screenshot shows the Azure Data Factory pipeline editor. The pipeline is named "pl_emr_src_to_landing" and contains a "ForEach1" loop. Inside the loop, there is a "Lookup" activity named "Lookup1" and a "Copy data" activity. The "Lookup1" activity is configured with the following settings:

- Source dataset: AzureDatabricksDeltaLakeDataset1
- Dataset properties:
 - schema_name: 'aa'
 - table_name: 'aa'
- First row only: ☒
- Use query: ☒ Query
- Query: @concat('select coalesce(cast(max(loaddate) as date),'', '1900-01-01', '') as last_fetched_date from audit.load_logs where', ' data_source=''', item().datasource, '' and tablename=''', item().tablename, ''')

The "Copy data" activity is also visible, with a "Full_load_cp" dataset selected. The "Pipeline expression builder" is open on the right, showing the query used in the "Lookup" activity.

Incremental load:

Source Path: bronze/hosa

The screenshot shows the Azure Data Factory pipeline editor. The pipeline is named "pl_emr_src_to_landing" and contains a "ForEach1" loop. Inside the loop, there is a "Copy data" activity named "Full_load_cp". The "Copy data" activity is configured with the following settings:

- Source dataset: generic_sql_ds
- Dataset properties:
 - db_name: @item().database
 - schema_name: @split(item().tablename, '.')[0]
 - table_name: @split(item().tablename, '.')[1]

The "Copy data" activity is also visible, with a "Full_load_cp" dataset selected. The "Source" tab is selected, showing the dataset properties.

Target Path for Data Loads: bronze/<target-path>

Copy data

Full_load_cp

General Source Sink Mapping Settings User properties

Sink dataset * generic_adls_parquet_ds

Dataset properties

Name	Value
file_name	@split(item().tablename, '.')[1]
file_path	@item().targetpath
container	bronze

Query: @concat('select *,'',item().datasource,'' as datasource from ',item().tablename,' where ',item().watermark,' >= ',activity('Fetch_logs').output.firstRow.last_fetched_date, ''')

Pipeline expression builder

Add dynamic content below using any combination of expressions, functions and system variables.

```
@concat('select *,'',item().datasource,'' as datasource from ',item().tablename,' where ',item().watermark,' >= ',activity('Fetch_logs').output.firstRow.last_fetched_date, ''')
```

Clear contents

Activity outputs Parameters System variables Functions Variables

Search

ArchiveFile
ArchiveFile activity output

Fetch_logs
Fetch_logs activity output

Fetch_logs first row
Data of the first row

OK Cancel

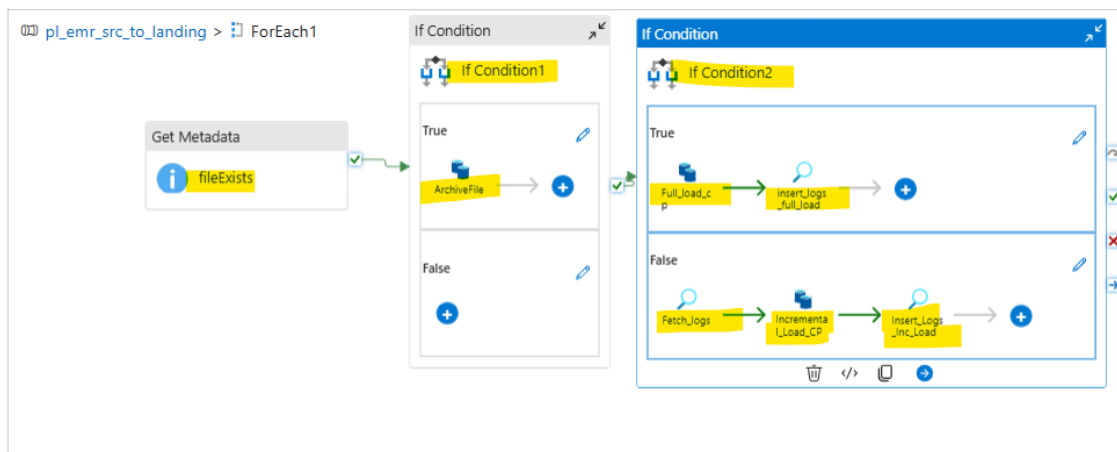
Lookup:

The screenshot displays the Microsoft Fabric pipeline editor. On the left, the pipeline canvas shows a sequence of activities: 'Fetch_logs', 'Copy data' (with 'Incremental_Load_CP' as a sub-activity), and 'Lookup2'. The 'Lookup2' activity is selected, and its configuration pane on the right is open. The 'Source dataset' is set to 'AzureDatabricksDeltaLakeDataset1'. Under 'Dataset properties', the 'Name' is 'aa', and the 'Value' is 'aa'. The 'First row only' checkbox is checked. The 'Query' field contains the expression: `@concat('insert into audit.load_logs(')`. The 'Pipeline expression builder' pane on the right shows the full expression: `@concat('insert into audit.load_logs(data_source,tablename, numberofrowscopied,watermarkcolumnname,loaddate) values ('', item().datasource,'', '',item().tablename,'', '',activity ('Incremental_Load_CP').output.rowscopied,'', '',item(). watermark,'', '',utcNow(),''))')`. Below the expression builder, the 'Activity outputs' tab is selected, showing a list of outputs: 'ArchiveFile', 'Fetch_logs', and 'Fetch_logs first row'.

This is our complete pipeline:

Before running the pipeline for each activity select the “sequential” option as shown below.

The screenshot shows the Microsoft Fabric pipeline editor with the 'ForEach' activity selected. The 'Activities' pane on the left lists various activity types. The 'ForEach' activity is highlighted, and its configuration pane is open. The 'Settings' tab is selected, and the 'Sequential' option is checked. The 'Items' field contains the expression: `@activity('lkp_EMR_configs').output....`. The pipeline canvas shows the 'ForEach' activity with a sub-activity 'lkp_EMR_configs'.



But limitation with this pipeline is it is sequential which will we resolve in part 2

Part 2:

In this section, we will focus on improving our data pipeline and governance.

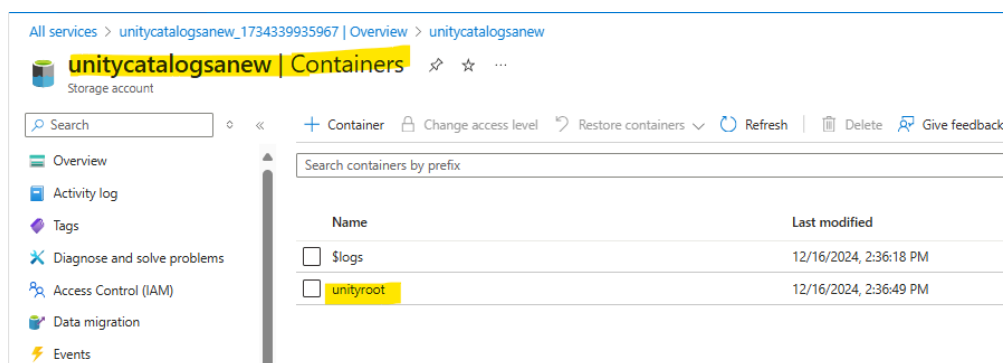
Transition from a local Hive Metastore to Databricks Unity Catalog for centralized metadata management and improved data governance.

We will first create a new databricks workspace “tt-hc-adb-ws”, select the “Premium (+ Role-based access controls)” while creating the workspace.

To configure the Metastore for the Unity catalog follow below steps:

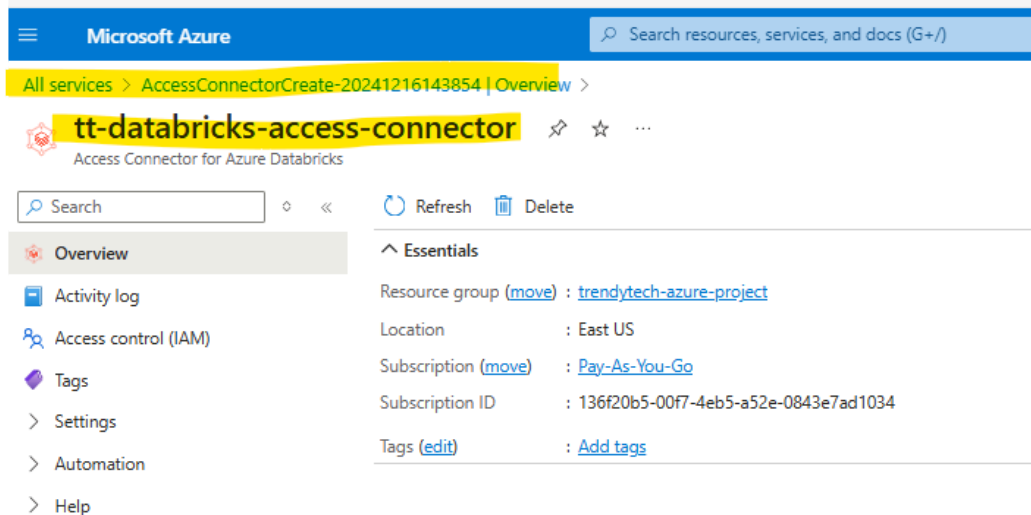
Step 1: Configure Storage for your Metastore - ADLS gen2 table -

Created a storage account for ADLS Gen2.- “ttunitycatalogsa”. Create a storage container that will hold your unity catalog metastore's metadata and managed data.
container - unityroot

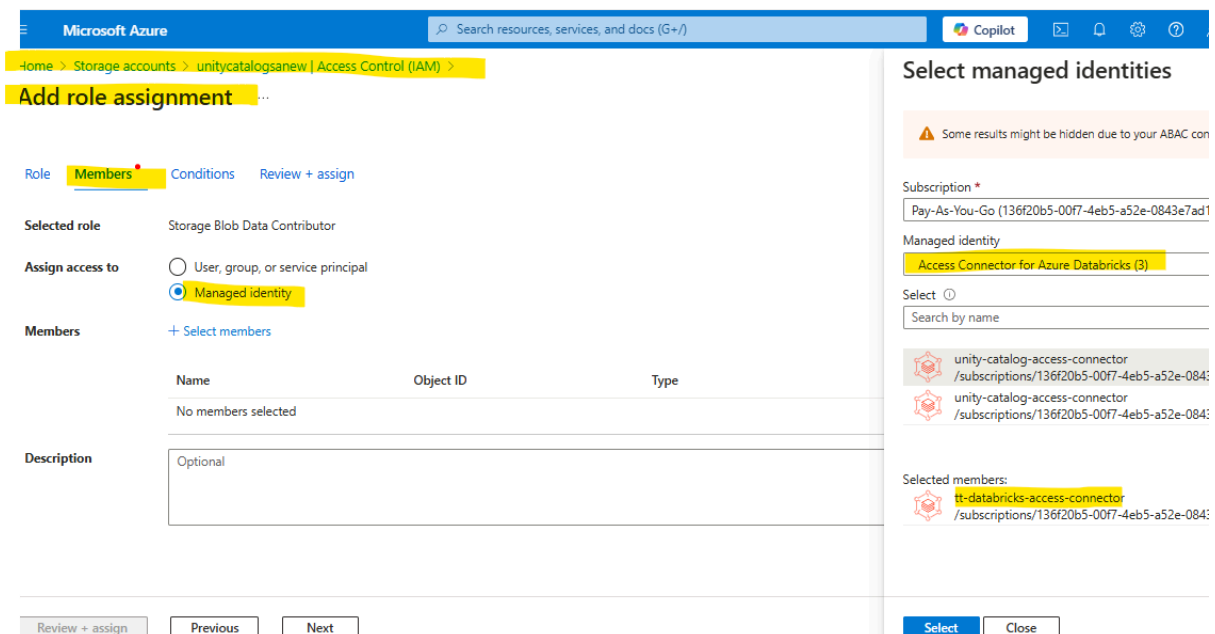


2. In Azure, create a databricks access connector that holds a managed identity and give it access to the storage container.

Go to “Access Connector for Azure Databricks” => create databricks access connector “tt-databricks-access-con”



Then Go to your storage account and click on access control (IAM)=> click on add -> role assignment -> storage blob data contributor -> in managed identity select the databricks connector that we have created.



Save it and then review and assign it.

Step 2: You will go to “https://accounts.azuredatabricks.net” in your browser. & there we will create the metastore.

Clicked on Catalog => create metastore -> region should be the same where you have the storage account and the workspaces -> storage account path:

Ex:

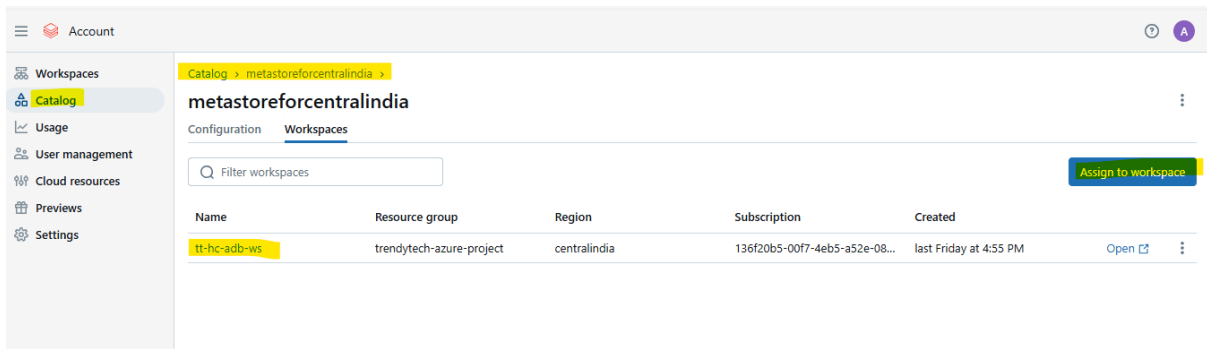
unityroot@ttunitycatalogsa.dfs.core.windows.net/

-> access connector id (mention the access connector id)

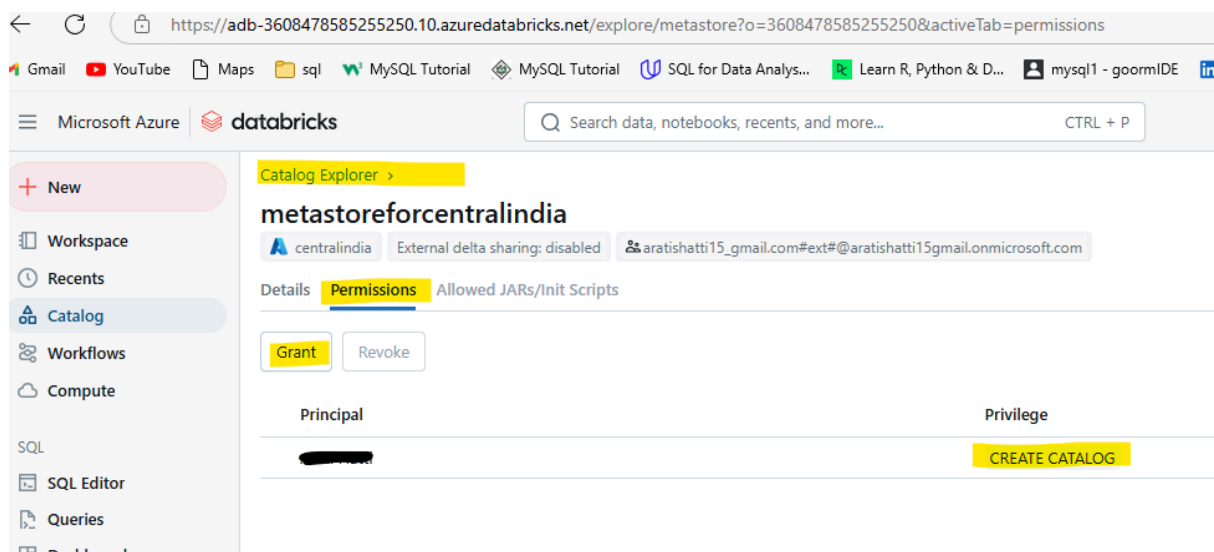
The screenshot shows the 'Create metastore' page in the Azure Databricks portal. The page is divided into a left sidebar with navigation links (Workspaces, Catalog, Usage, User management, Cloud resources, Previews, Settings) and a main content area. The main content area has a breadcrumb 'Catalog > Create metastore' and a progress indicator showing two steps: '1 Create metastore' and '2 Assign to workspaces'. The 'Name' field is filled with 'MetastoreForCentralIndia'. The 'Region' dropdown is set to 'centralindia'. The 'ADLS Gen 2 path (optional)' field is filled with 'unityroot@unitycatalogsanew.dfs.core.windows.net/'. The 'Access Connector Id' field is filled with '/subscriptions/136f20b5-00f7-4eb5-a52e-0843e7ad1034/resourceGroups/trendytech-azure-project/'. There is an 'Advanced options' button at the bottom.

Also we will assign this metastore to our workspace.

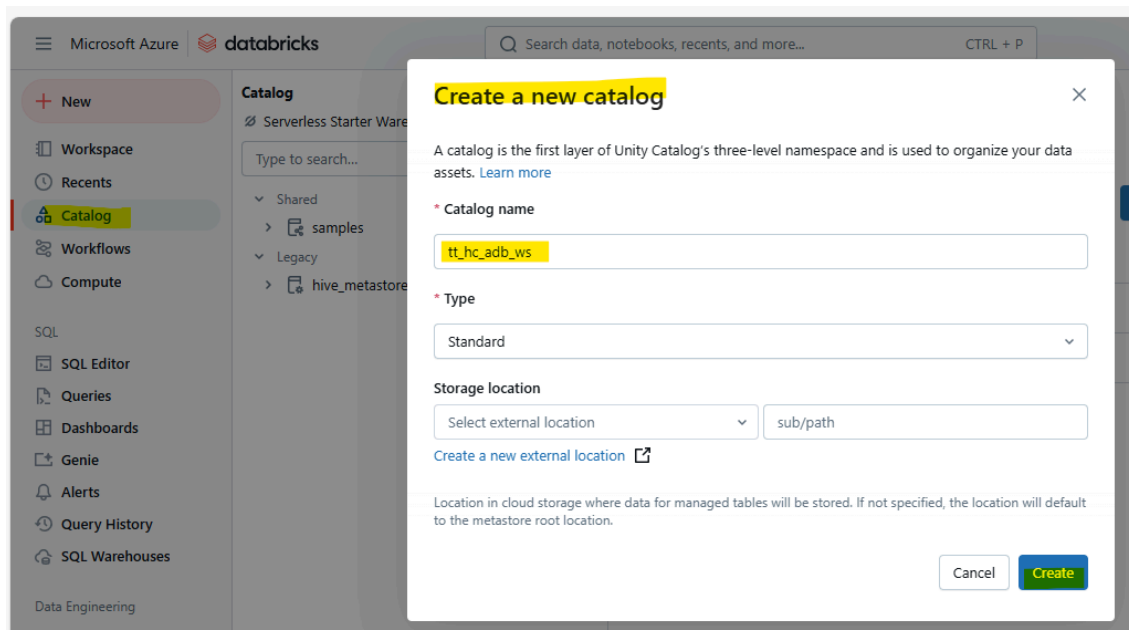
Go to Catalog => select the recently created metastore => click on Assign to workspace => select the workspace that you want to assign.

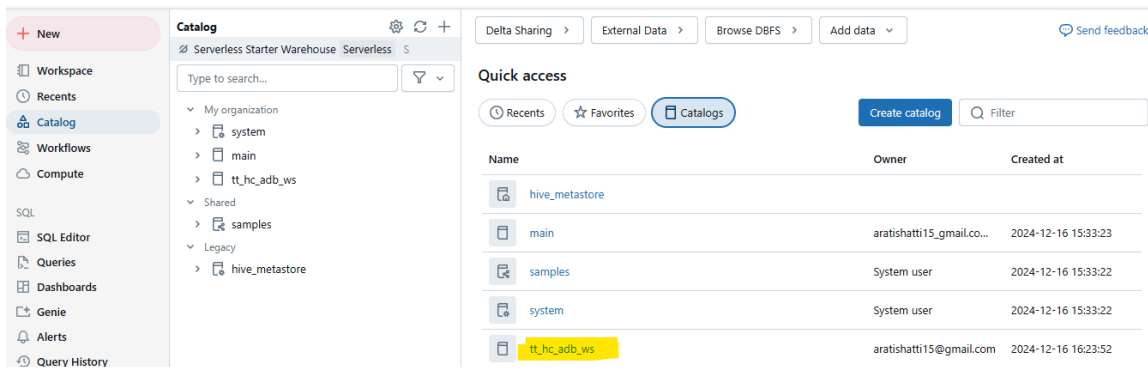


And we will assign “create catalog” permission to the user for this metastore if you are not getting the option to create a new Catalog.



Now we will create the catalog “tt_hc_adb_ws”.





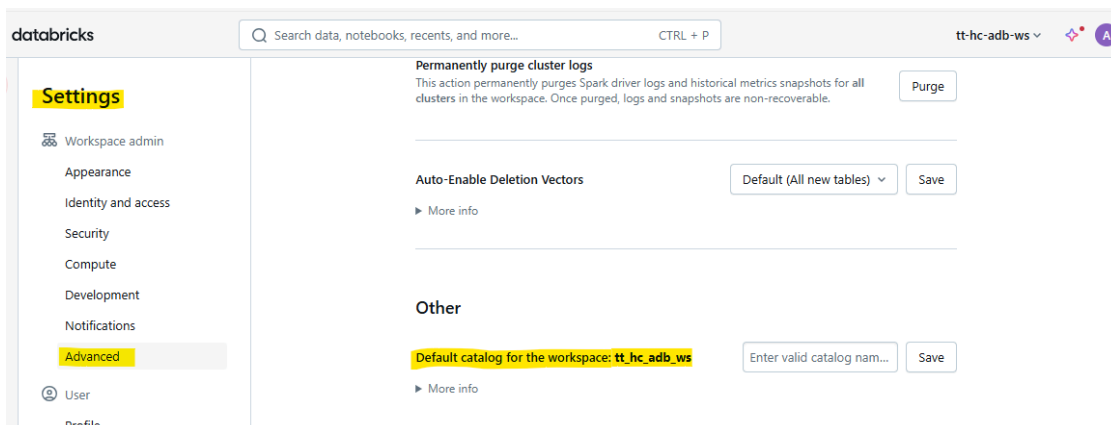
After creating this catalogue, set this catalogue as a default catalog for the workspace “tt-hc-adb-ws”. For this follow below steps:

Open your Databricks Workspace.

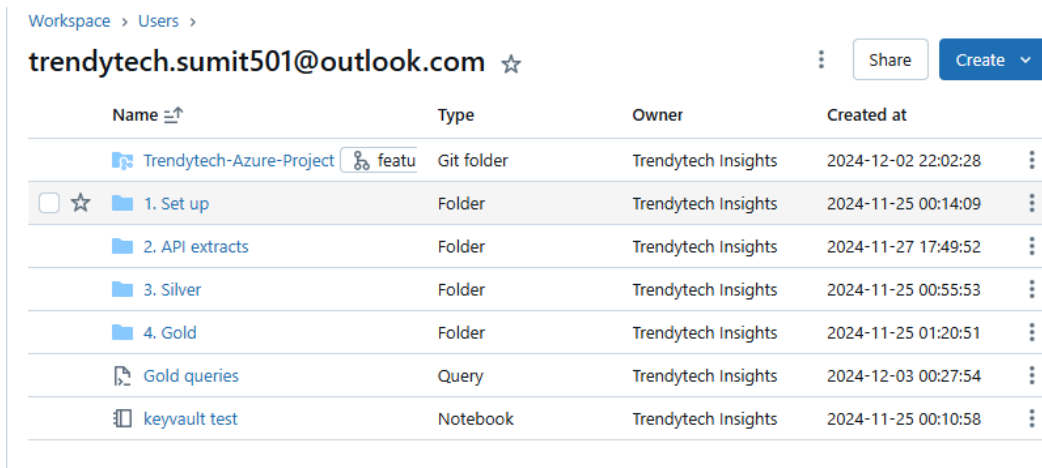
Navigate to Admin Settings → Workspace Settings.

Set the Default Catalog to your desired catalog (e.g., tt_hc_adb_ws).

Refer below screenshot for more clarity.



Also to organize the notebook we will create the folder as shown below.



1. Set up:

Workspace > Users > trendytech.sumit501@outlook.com >

1. Set up ☆

Share Create

Name	Type	Owner	Created at
1. audit_ddl	Notebook	Trendytech Insights	2024-11-25 00:14:14
2. adls_mount	Notebook	Trendytech Insights	2024-11-25 00:16:49

2. API extracts

Workspace > Users > trendytech.sumit501@outlook.com >

2. API extracts ☆

Share Create

Name	Type	Owner	Created at
ICD Code API extract	Notebook	Trendytech Insights	2024-11-28 08:57:55
NPI API extract	Notebook	Trendytech Insights	2024-11-27 17:50:07

3. Silver

Workspace > Users > trendytech.sumit501@outlook.com >

3. Silver ☆

Share Create

Name	Type	Owner	Created at
Claims	Notebook	Trendytech Insights	2024-11-25 00:55:54
CPT codes	Notebook	Trendytech Insights	2024-11-28 23:13:09
Departments_F	Notebook	Trendytech Insights	2024-11-25 00:55:55
Encounters	Notebook	Trendytech Insights	2024-11-25 00:55:55
ICD Code	Notebook	Trendytech Insights	2024-11-28 09:08:35
NPI	Notebook	Trendytech Insights	2024-11-26 14:42:54
Patient	Notebook	Trendytech Insights	2024-11-25 00:55:54
Providers_F	Notebook	Trendytech Insights	2024-11-25 00:55:55
Transactions	Notebook	Trendytech Insights	2024-11-25 00:55:53

4. Gold

Workspace > Users > trendytech.sumit501@outlook.com >

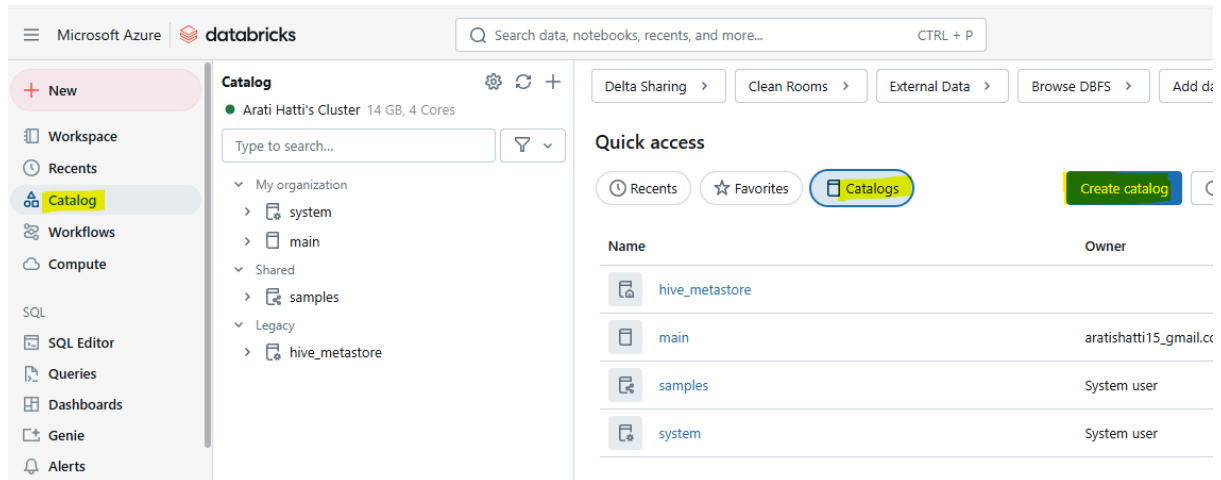
4. Gold ☆

Share Create

Name	Type	Owner	Created at
dim_cpt_code	Notebook	Trendytech Insights	2024-11-28 23:36:42
dim_department	Notebook	Trendytech Insights	2024-11-25 01:42:29
dim_icd_code	Notebook	Trendytech Insights	2024-11-28 09:11:38
dim_npi	Notebook	Trendytech Insights	2024-11-28 09:15:54
dim_patient	Notebook	Trendytech Insights	2024-11-25 01:20:59
dim_provider	Notebook	Trendytech Insights	2024-11-25 01:29:07
fact_transaction	Notebook	Trendytech Insights	2024-11-26 14:22:04

Note: After creating the databricks workspace, enable the DBFS.

Create the catalog “tt-hc-adb-ws” as shown below.



We will create the audit database using as shown below



Note: Before proceeding ahead we will create the key vault in azure and will store key for our ADLS gen2 storage in it. Refer below steps for more clarification.

Key Vault creation in Azure:

Steps to Store Key in Key Vault and Create Secret Scope in Databricks:

1. Store the Storage Account Access Key in Azure Key Vault

Create an Azure Key Vault (if not already created):

Go to the Azure Portal. => Search for "Key Vault" and click Create.=> Fill in the necessary details (Resource Group, Key Vault Name, Region) and click Review + Create.
Refer below screenshot for more details.

The screenshot shows the 'Create a key vault' page in the Azure Portal. The breadcrumb navigation is 'Home > Key vaults >'. The main heading is 'Create a key vault'. Below this, the 'Project details' section is highlighted. It includes a description: 'Select the subscription to manage deployed resources and costs. Use resource groups like folders to organize and manage all your resources.' The form fields are: 'Subscription *' with a dropdown showing 'Pay-As-You-Go (136f20b5-00f7-4eb5-a52e-0843e7ad1034)'; 'Resource group *' with a dropdown showing 'trendytech-azure-project' and a 'Create new' link; 'Instance details' section with 'Key vault name *' (dropdown showing 'kv-heathcare-project' with a green checkmark), 'Region *' (dropdown showing 'Central India'), and 'Pricing tier *' (dropdown showing 'Standard'). The 'Recovery options' section is partially visible at the bottom.

In Access configuration select option vault access policy option and select define permission for your username.

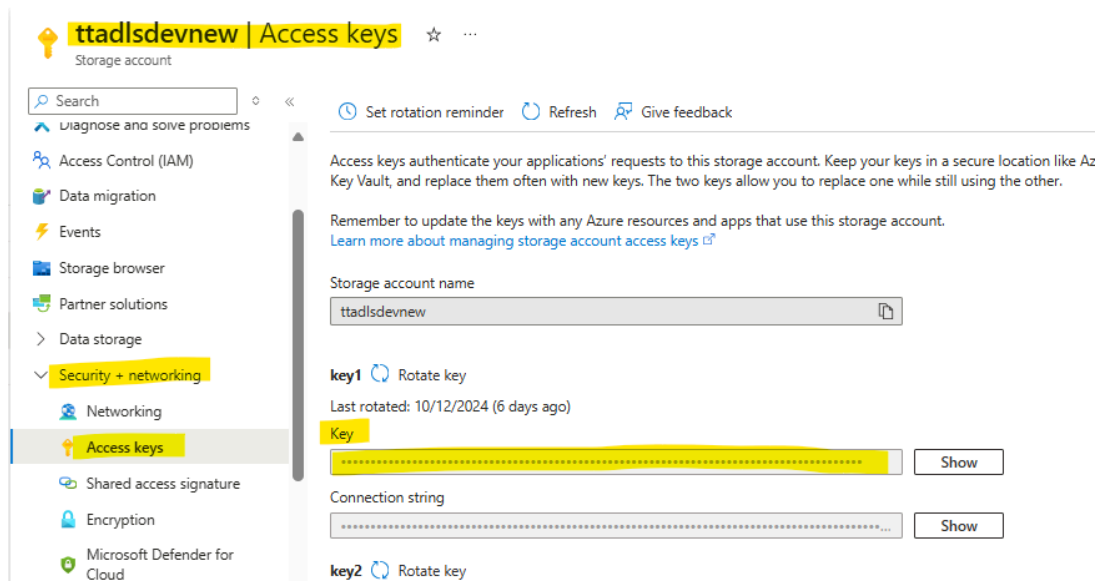
The screenshot shows the 'Access configuration' section of the 'Create a key vault' page. It includes a description: 'Configure data plane access for this key vault. To access a key vault in data plane, all callers (users or applications) must have proper authentication and authorization. Authentication establishes the identity of the caller. Authorization determines the permissions granted to the caller.' The 'Permission model' section has two options: 'Azure role-based access control (recommended)' and 'Vault access policy' (selected with a blue dot). The 'Resource access' section has three checkboxes: 'Azure Virtual Machines for deployment', 'Azure Resource Manager for template deployment', and 'Azure Disk Encryption for volume encryption'. The 'Access policies' section has a description: 'Access policies enable you to have fine grained control over access to vault items. Learn more'. Below this is a table with columns: 'Name', 'Email', 'Key Permissions', and 'Secret Permissions'. The table has one row for a user named 'USER' with email 'aratishatti15_gmail.com#EXT#@aratishatti15gmail.o...'. The 'Key Permissions' column shows 'Get, List, Update, Create, Import, Delete, Recover, Ba...' and the 'Secret Permissions' column shows 'Get, List, Set, Delete'. At the bottom are buttons for 'Previous', 'Next', and 'Review + create'.

Then create the key vault.

Add the Storage Account Access Key to the Key Vault, to add follow below steps:

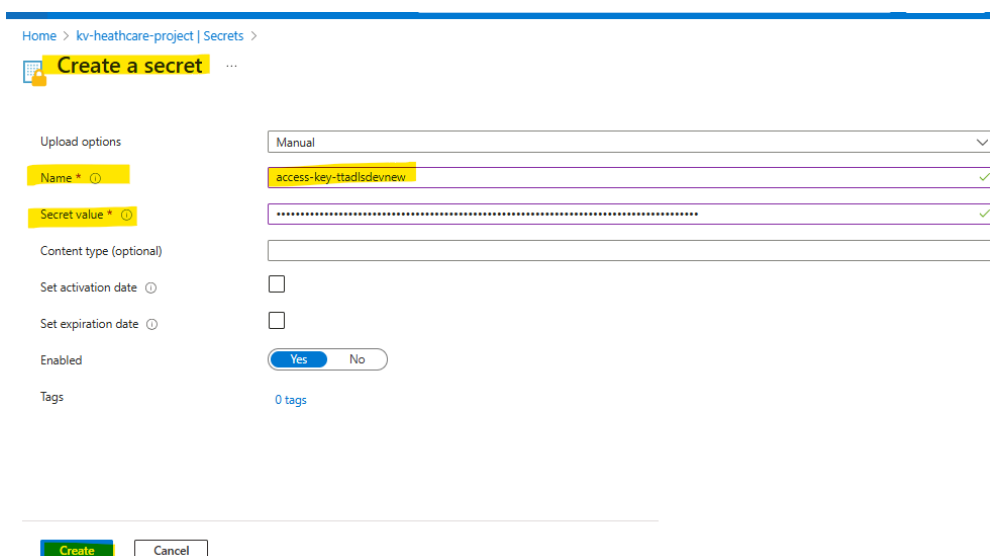
First get the storage account key, to get the storage account access key follow below steps:

=> Go to your Azure Storage Account => Security & Networking> Access Keys.
=> Copy the key under the "Key1" or "Key2" section.



To store keys in the key vault, navigate to your Key Vault.

Click Secrets > + Generate/Import. => Enter the following details:
Name: Provide a descriptive name for the secret (e.g., adls-access-key).
Value: Copy the access key of your storage account and paste it here.
=> Click Create.



2. Assign Key Vault Access to Databricks

To allow Databricks to access the Key Vault:

Go to the Key Vault => Navigate to your Key Vault in the Azure Portal. => Click on Access Policies > + Add Access Policy. => Grant Permissions:

Template: Select Secret Management.

Home > kv-healthcare-project | Access policies >

Create an access policy

kv-healthcare-project

1 Permissions 2 Principal 3 Application (optional) 4 Review + create

Configure from a template

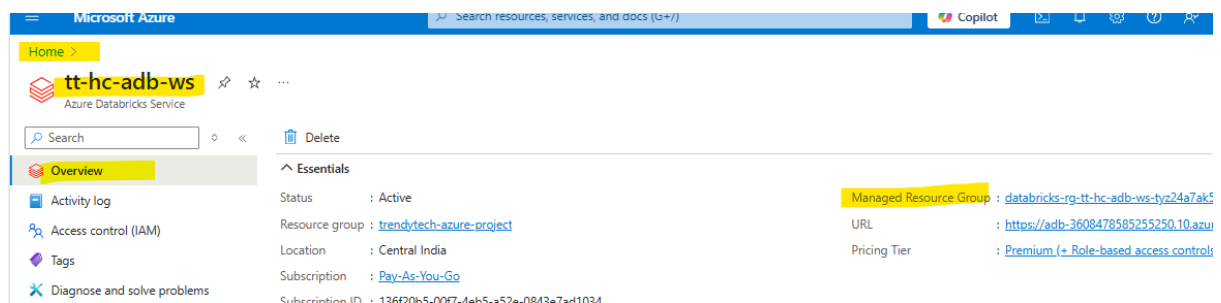
Key, Secret, & Certificate Management

Key permissions	Secret permissions	Certificate permissions
Key Management Operations	Secret Management Operations	Certificate Management Operations
☒ Select all	☒ Select all	☒ Select all
☒ Get	☒ Get	☒ Get
☒ List	☒ List	☒ List
☒ Update	☒ Set	☒ Update
☒ Create	☒ Delete	☒ Create
☒ Import	☒ Recover	☒ Import
☒ Delete	☒ Backup	☒ Delete
☒ Recover	☒ Restore	☒ Recover

Previous Next

Principal: Add your Azure Databricks Managed Identity.

You can find the Managed Identity from the Databricks Resource in Azure Portal under Managed Identity.



Now select this managed resource group and check the managed identity and define policy for it.

Home > tt-hc-adb-ws >

databricks-rg-tt-hc-adb-ws-tyz24a7ak54oe ☆ ☆ ...

Resource group

Search

+ Create Manage view Delete resource group Refresh Export to CSV Open query Assign

Overview

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
- Cost Management
- Monitoring

Essentials

Resources Recommendations (1)

Filter for any field... Type equals all Location equals all Add filter

Showing 1 to 5 of 5 records. Show hidden types

Name ↑↓	Type ↑↓
dbmanagedidentity	Managed Identity
dbstoragesonxvyaont3lc	Storage account

Save the Policy:

Home > kv-heathcare-project | Access policies >

Create an access policy

kv-heathcare-project

Permissions 2 Principal 3 Application (optional) 4 Review + create

Only 1 principal can be assigned per access policy.
Use the new embedded experience to select a principal. The previous popup experience can be accessed here. [Select a principal](#)

dbmanagedidentity

dbmanagedidentity

Selected item

No item selected

Previous Next

Click Add, then Save the changes.

3. Create a Secret Scope in Databricks

Once the secret is stored in Key Vault, create a secret scope in Databricks.

Open Databricks Workspace:

Go to your Azure Databricks Workspace.

While creating the secret scope, edit the url till .net and add “#secrets/createScope”

Create a Secret Scope Linked to Key Vault:

Enter the following:

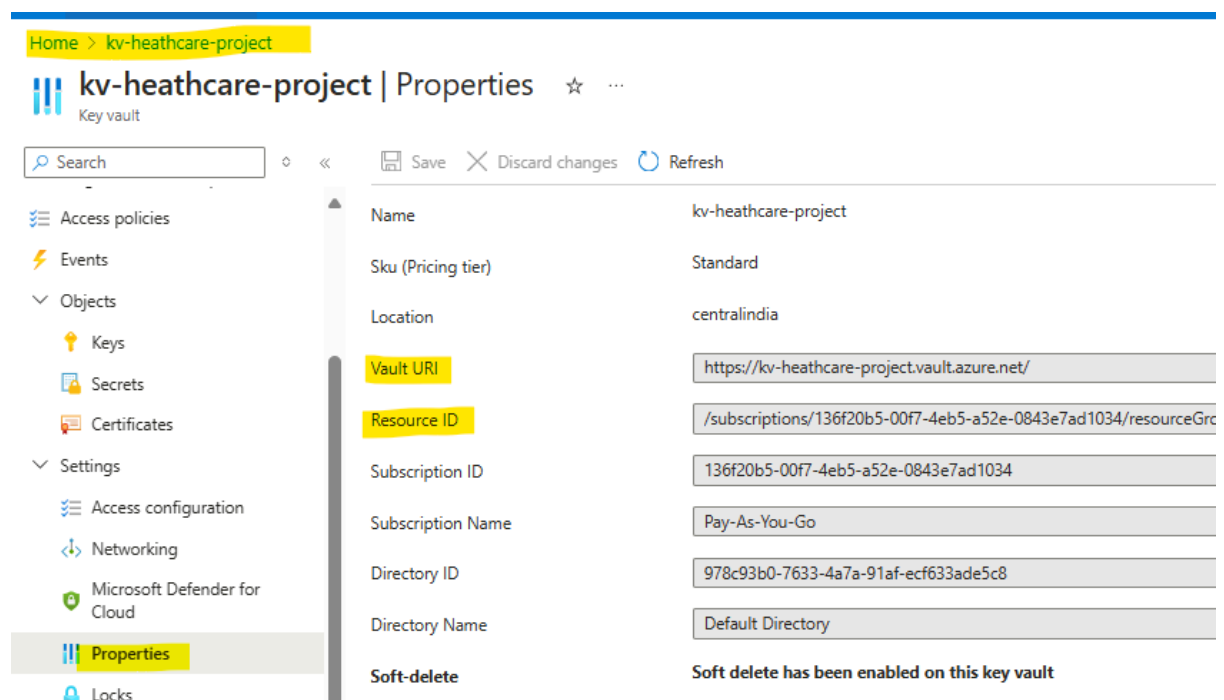
Scope Name: Give a name to the scope (e.g., adlsdevnew).

Scope Backing: Select Azure Key Vault.

DNS Name: Enter the Key Vault URL.

To get these details go to key vault => Setting => Properties

Refer below screenshot for more clarity.



Now provide these details in scope creation and create the scope.

HomePage / Create Secret Scope

Create Secret Scope

A store for secrets that is identified by a name and backed by a specific store type. [Learn more](#)

Scope Name [?]

tt-hc-kv

Manage Principal [?]

Creator

Azure Key Vault [?]

DNS Name

https://kv-heathcare-project.vault.azure.net/

Resource ID

/subscriptions/136f20b5-00f7-4eb5-a52e-0843e7ad1034/resourceGroups/trendytec

Verify the Secret Scope:

Similarly first we store credentials for sql database (Username and password) and databricks also (In case of Databricks we will store access token).

To get access token in Databricks got to Setting => Developer => Access token => Generate Access token

kv-heathcare-project | Secrets

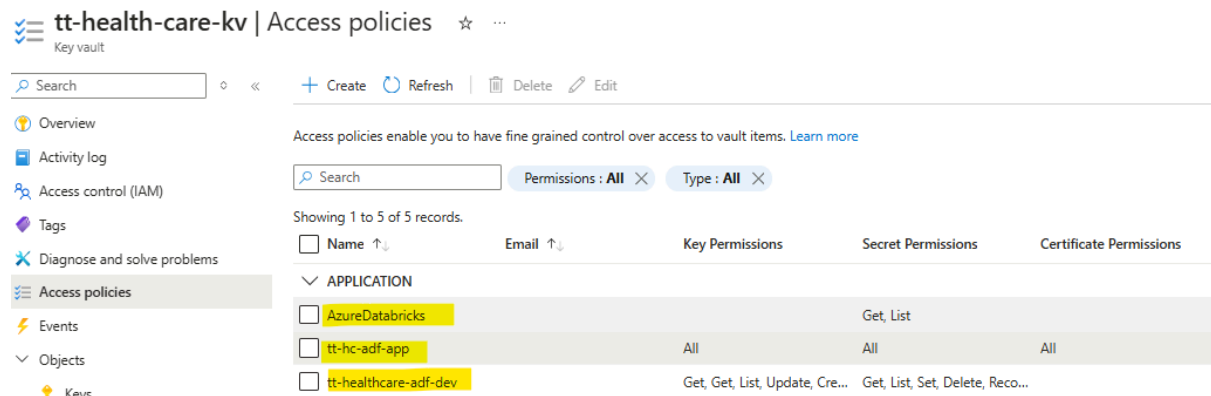
Overview Activity log Access control (IAM) Tags Diagnose and solve problems Access policies Events Objects Keys

Generate/Import Refresh Restore Backup View sample code Manage deleted secrets

The secret 'db-hc-adb-ws-at' has been successfully created.

Name	Type	Status	Expiration date
db-hc-adb-ws-at		✓ Enabled	
azure-sql-db-password		✓ Enabled	
azure-sql-db-username		✓ Enabled	
access-key-ttadlsdevnew		✓ Enabled	

Now for ADF and ADLS gen we will create application and will provide permission (You can Provide all the permission) as shown by Sumit Sir in the video



tt-health-care-kv | Access policies ☆ ...

Key vault

Search

+ Create Refresh Delete Edit

Overview

Activity log

Access control (IAM)

Tags

Diagnose and solve problems

Access policies

Events

Objects

Keys

Access policies enable you to have fine grained control over access to vault items. [Learn more](#)

Search

Permissions : All Type : All

Showing 1 to 5 of 5 records.

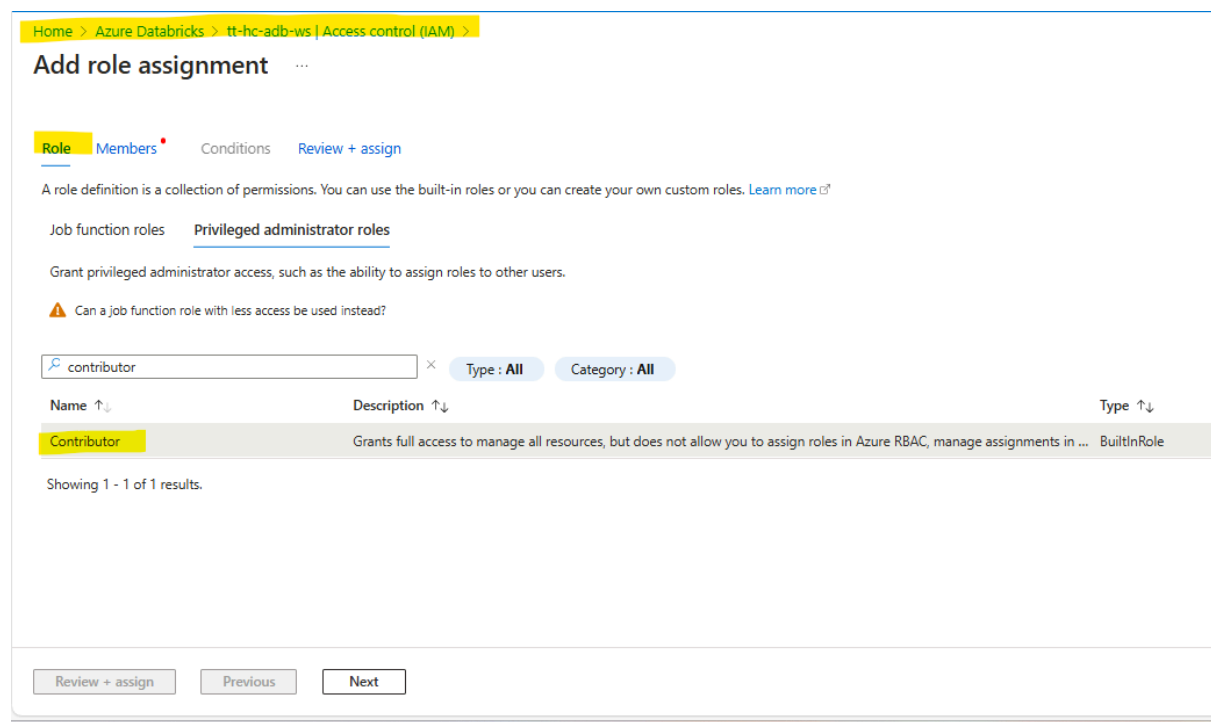
Name	Email	Key Permissions	Secret Permissions	Certificate Permissions
APPLICATION				
AzureDatabricks			Get, List	
tt-hc-adf-app		All	All	All
tt-healthcare-adf-dev		Get, Get, List, Update, Cre...	Get, List, Set, Delete, Reco...	

To perform role assignments in Azure Databricks and Azure Data Factory (ADF), follow the steps outlined below.

For Azure Databricks:

To assign the role : Go to resource => IAM => Add => Role Assignment.

Define Contributor role for the application that is created for databricks as shown below.



Home > Azure Databricks > tt-hc-adb-ws | Access control (IAM)

Add role assignment

Role Members Conditions Review + assign

A role definition is a collection of permissions. You can use the built-in roles or you can create your own custom roles. [Learn more](#)

Job function roles Privileged administrator roles

Grant privileged administrator access, such as the ability to assign roles to other users.

⚠ Can a job function role with less access be used instead?

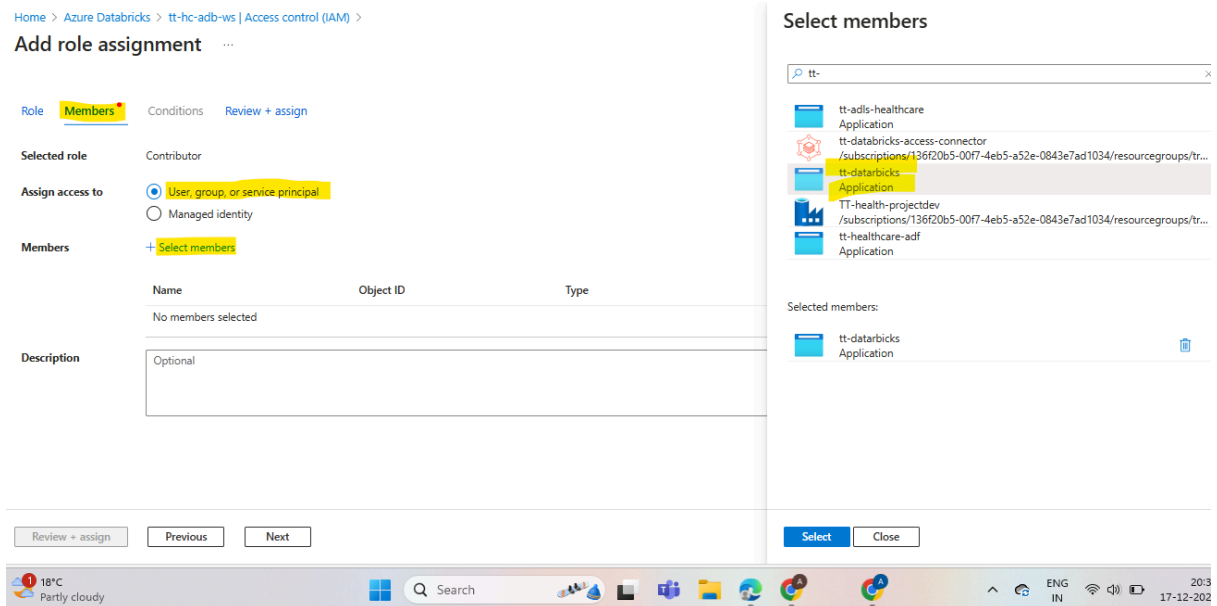
contributor

Type : All Category : All

Name	Description	Type
Contributor	Grants full access to manage all resources, but does not allow you to assign roles in Azure RBAC, manage assignments in ...	BuiltInRole

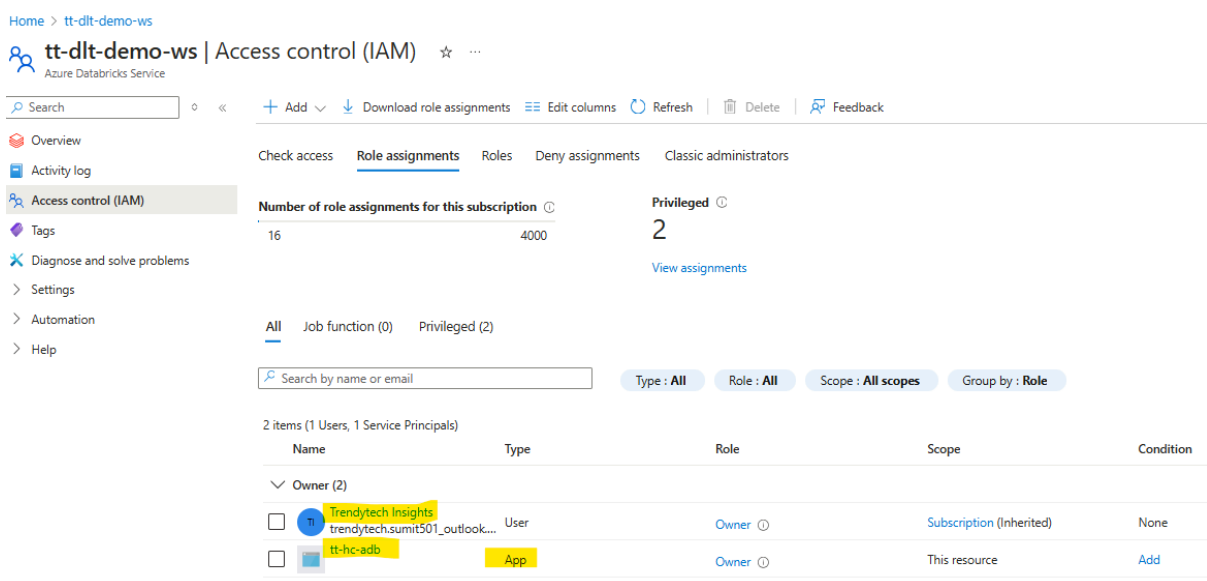
Showing 1 - 1 of 1 results.

Review + assign Previous Next



Note: In condition give select “Allow user to assign all roles (highly privileged)”

Also for your username define the owner role if it is not present as shown below.



For ADF:

To assign the role : Go to resource => IAM => Add => Role Assignment.

Define Contributor role for the application that is created for databricks as shown below.

Home > tt-healthcare-adf-dev

tt-healthcare-adf-dev | Access control (IAM) ☆ ...
Data factory (V2)

Search ○ ◁ + Add ▾ ↓ Download role assignments ≡ Edit columns ↻ Refresh | 🗑 Delete | 🗨 Feedback

Overview
Activity log
Access control (IAM)
Tags
Diagnose and solve problems
Settings
Getting started
Monitoring
Automation
Help

Number of role assignments for this subscription ⓘ
16 4000

Privileged ⓘ
3
[View assignments](#)

All Job function (0) Privileged (3)

Search by name or email

Type: All Role: All Scope: All scopes Group by: Role

3 items (1 Users, 2 Service Principals)

Name	Type	Role	Scope
▼ Owner (3)			
☐ Trendytech Insights trendytech.sumit501_outlook...	User	Owner ⓘ	Subscription (Inherited)
☐ tt-adf-healthcare-app	App	Owner ⓘ	This resource
☐ tt-hc-adf-app	App	Owner ⓘ	This resource

We will use the code in the adls_mount notebook to mount the containers in our storage account.

```

storageAccountName = "ttadlsdevnew"
storageAccountAccessKey = dbutils.secrets.get('tt-hc-kv', 'access-key-ttadlsdevnew')
mountPoints=["gold","silver","bronze","landing","configs"]
for mountPoint in mountPoints:
    if not any(mount.mountPoint == f"/mnt/{mountPoint}" for mount in dbutils.fs.mounts()):
        try:
            dbutils.fs.mount(
                source = "wasbs://{}/@{}.blob.core.windows.net".format(mountPoint, storageAccountName),
                mount_point = f"/mnt/{mountPoint}",
                extra_configs = {'fs.azure.account.key.' + storageAccountName + '.blob.core.windows.net': storageAccountAccessKey}
            )
            print(f"{mountPoint} mount succeeded!")
        except Exception as e:
            print("mount exception", e)

```

gold mount succeeded!
silver mount succeeded!
bronze mount succeeded!
landing mount succeeded!
configs mount succeeded!

Now we will create a linked service for key vault and Databricks as shown below.

New linked service

Azure Key Vault

Name *

tt_hc_kv_ls

Description

Azure key vault selection method ⓘ

☒ From Azure subscription ☐ Enter manually

Azure subscription ⓘ

Pay-As-You-Go (136f20b5-00f7-4eb5-a52e-0843e7ad1034) ▾

Azure key vault name *

kv-heathcare-project ▾

[Edit key vault](#)

Authentication method

System-assigned managed identity ▾

Managed identity name: TT-health-projectdev

Managed identity object ID: 8d35fbeb-2451-4cce-9d24-31bf7d57b55d

Grant Data Factory service managed identity access to your Azure Key Vault [Learn more](#) ↗

Create

Back

✔ Connection successful

Test connection

Cancel

Before creating the linked services, grant access permissions to the Data Factory service principal under the access policies in Key Vault and provide key and secret access permission.

+ Create

↻ Refresh

🗑 Delete

✎ Edit

Access policies enable you to have fine grained control over access to vault items. [Learn more](#)

🔍 Search

Permissions : All ×

Type : All ×

Showing 1 to 6 of 6 records.

☐ Name ↑↓	Email ↑↓	Key Permissions	Secret Permissions	Certificate Permissions
APPLICATION				
☐ AzureDatabricks			Get, List	
☐ dbmanagedidentity		Get, List, Update, Create, Import, Del...	Get, List, Set, Delete, Recover, Backup...	Get, List, Update, Create, Import, D
☐ tt-adls-healthcare		Get, List, Update, Create, Import, Del...	Get, List, Set, Delete, Recover, Backup...	Get, List, Update, Create, Import, D
☐ TT-health-projectdev		Get, List, Update, Create, Import, Del...	Get, List, Set, Delete, Recover, Backup...	
☐ tt-healthcare-adf		Get, List, Update, Create, Import, Del...	Get, List, Set, Delete, Recover, Backup...	Get, List, Update, Create, Import, D
USER				
☐ Arati Hatti	aratishatti15_gmail.com#EXT#@arati...	Get, List, Update, Create, Import, Del...	Get, List, Set, Delete, Recover, Backup...	Get, List, Update, Create, Import, D

After this you will be able to access secrets while creating linked services.

Note: While creating Linked service for Databricks select the existing cluster option as shown below.

Edit linked service

 Azure Databricks [Learn more](#)

Authentication type *

Access Token

Access token

Azure Key Vault

AKV linked service * ⓘ

tt_hc_kv_ls

Secret name *

db-hc-adb-ws-at

☒ Edit

Secret version ⓘ

Latest version

☐ Edit

Select cluster

☐ New job cluster
☒ Existing interactive cluster
☐ Existing instance pool

Choose from existing clusters * ⓘ

Sumit's Personal Compute Cluster

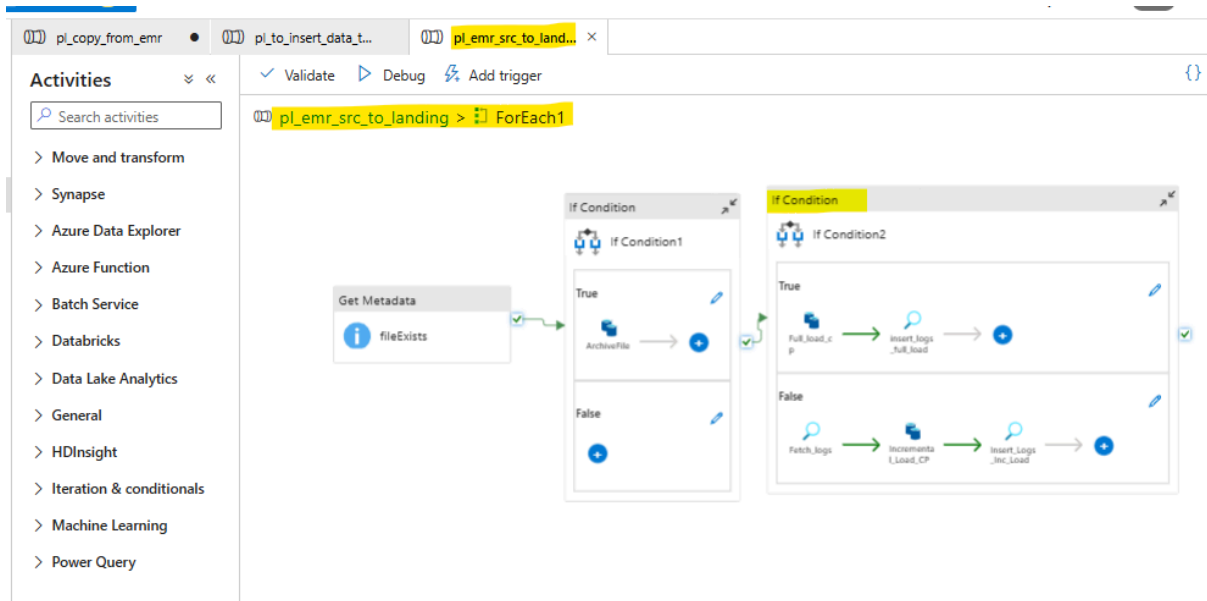
Annotations

We will update the linked services for ADLS Gen2, sql database and now instead of providing the storage account key directly, we will select the secret scope for respective ADLS gen storage, SQL DB, etc. as shown by Sumit sir in the video.

How to implement active and inactive Flag.

1. Creating new pipeline - pl_copy_from_emr

To create the pipeline “pl_copy_from_emr” and we will copy the “2nd If condition (having condition `@equals(item().loadtype, 'Full')`)” for copying the data into this pipeline.



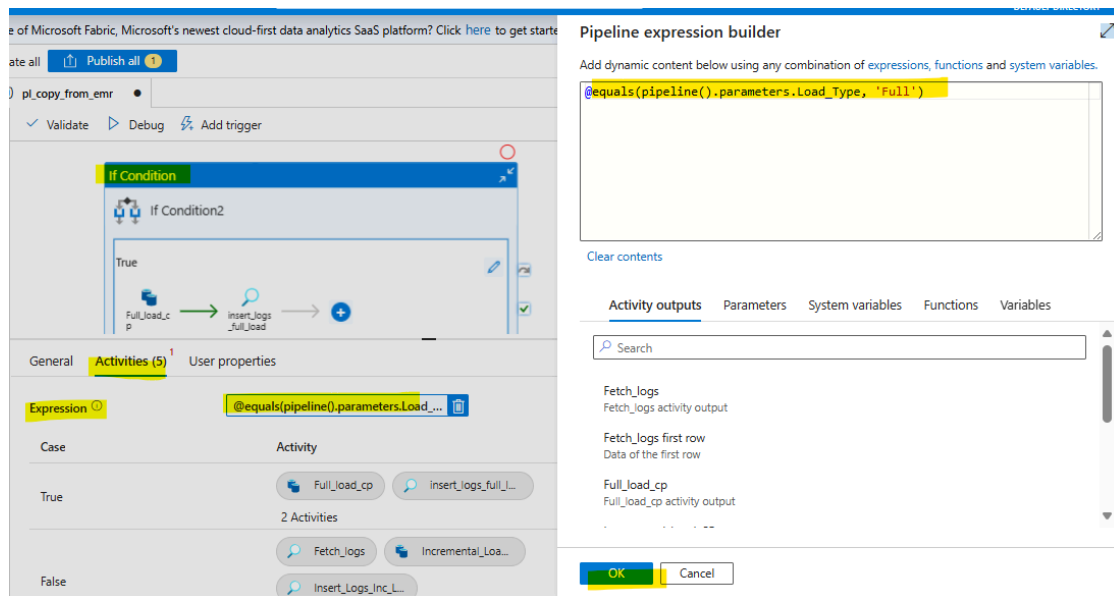
In the pipeline `pl_emr_src_to_landing`, when using `@equals(item().loadtype, 'Full')`, it works because the copy activity is directly inside the For Each loop, and `item()` refers to the current element being processed during the loop's iteration.

However, when you move the copy activity to a child pipeline (triggered using the Execute Pipeline activity) and place the If condition in the parent pipeline's For Each loop, the child pipeline no longer has direct access to `item()`. Instead, you need to pass the loop data (e.g., `loadtype`) as a parameter to the child pipeline.

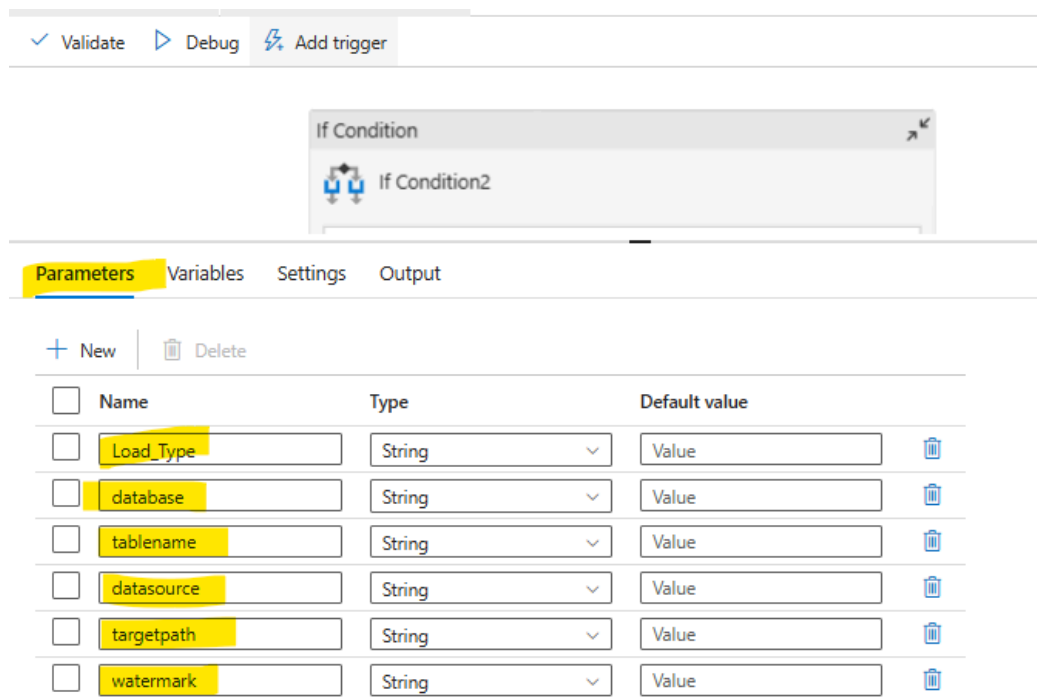
This is where `@equals(pipeline().parameters.Load_Type, 'Full')` comes into play. The parameter `Load_Type` is explicitly passed to the child pipeline when it's invoked, and the condition uses this parameter to determine the action. By referencing the parameter, the child pipeline can work independently while still receiving the necessary context from the parent pipeline.

So, in this pipeline update the Expression as shown below and also we have to add the parameter as shown below

```
@equals(pipeline().parameters.Load_Type, 'Full' )
```



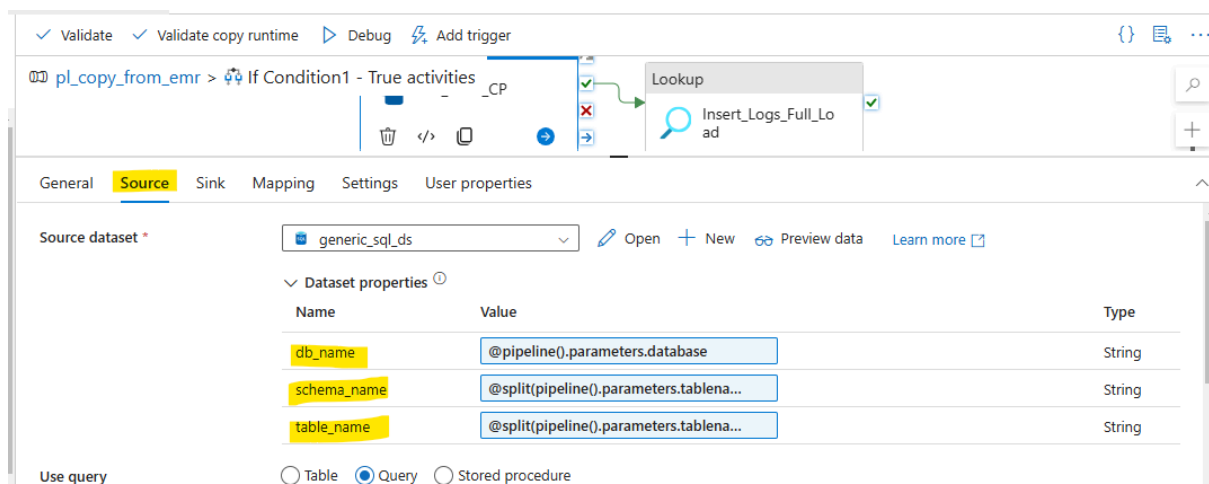
Parameter:



We will update the values for the source and sink variables for both the full load and incremental load copy activities. Since the variable values are the same for both, we will use the same variable, as demonstrated below.

Source:

```
db_name - @pipeline().parameters.database
schema_name - @split(pipeline().parameters.tablename, '.') [0]
table_name - @split(pipeline().parameters.tablename, '.') [1]
```

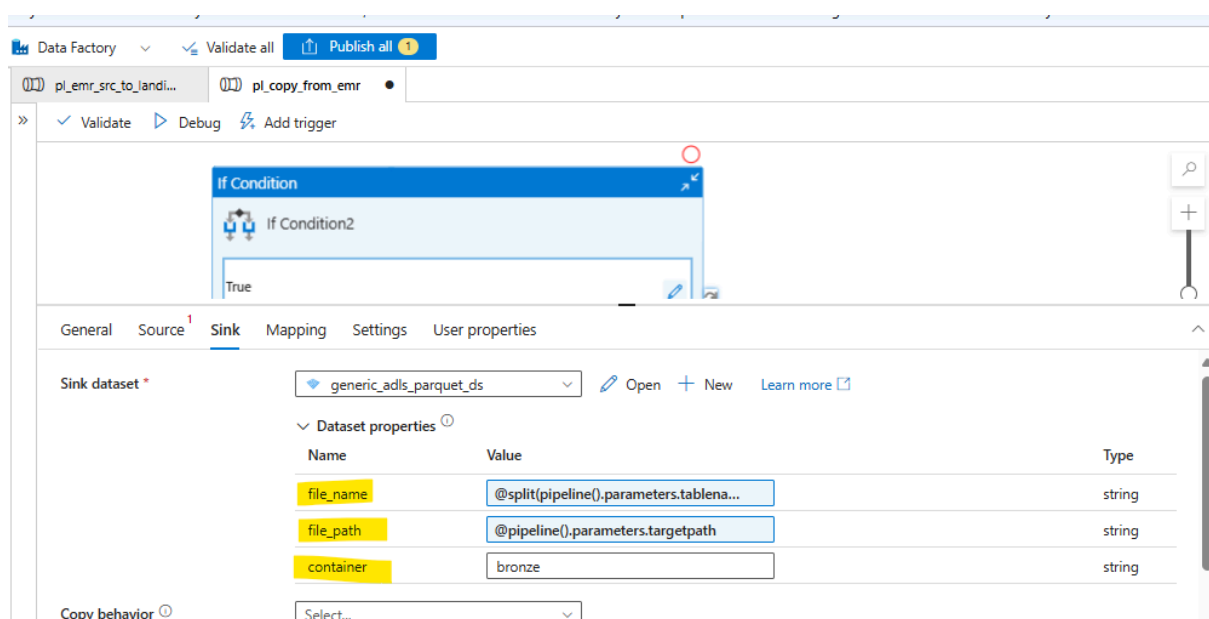


Sink:

```
file_name - @split(pipeline().parameters.tablename, '.')[1]
```

```
file_path - @pipeline().parameters.targetpath
```

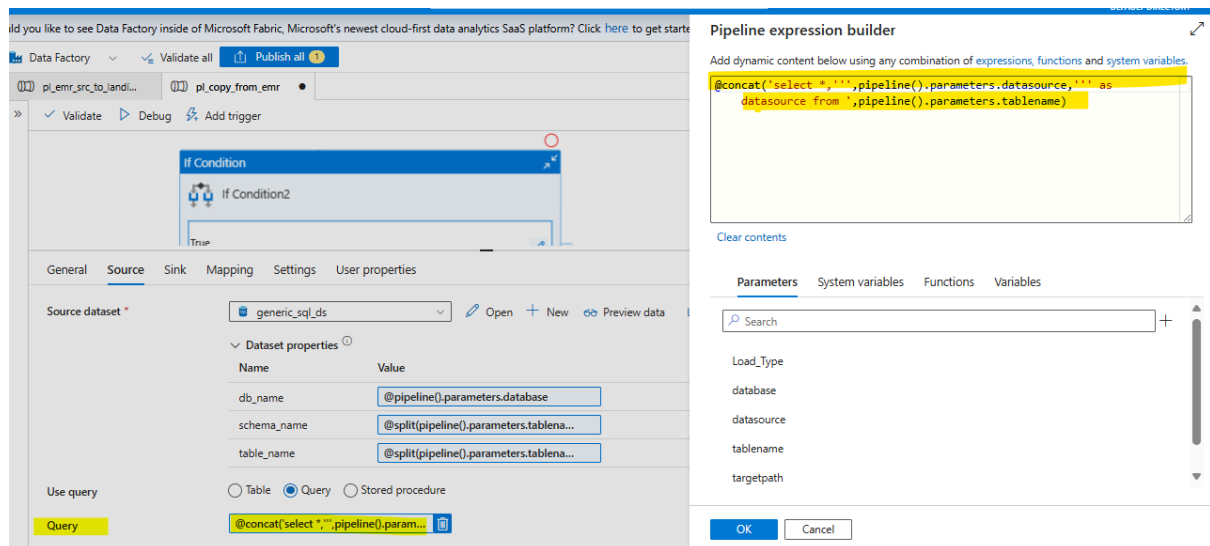
Container -bronze



Also we have to update the queries for Full load as shown below

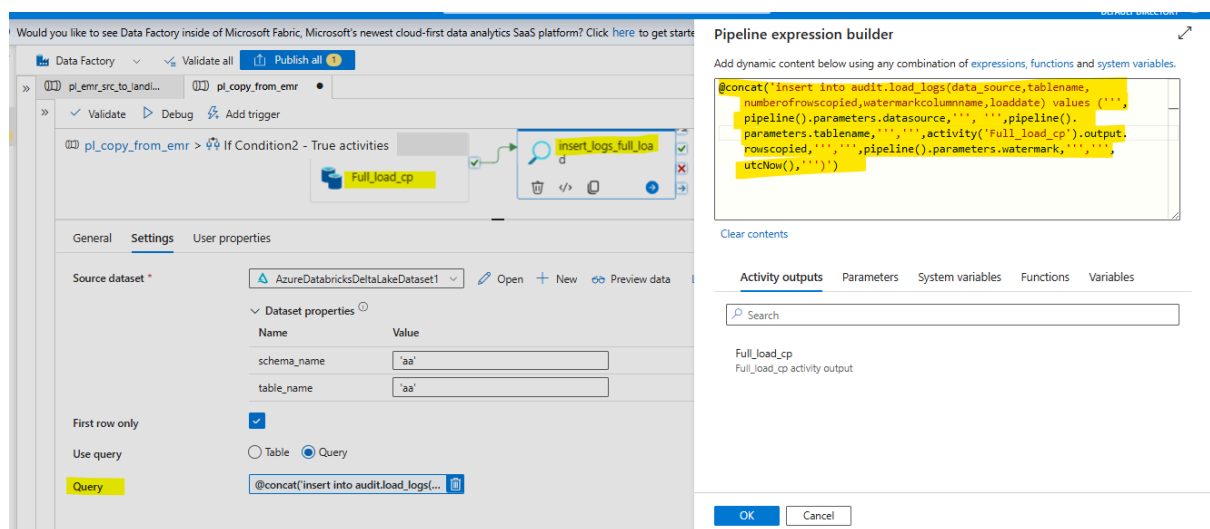
Full Load:

```
@concat('select *,'',pipeline().parameters.datasource, '' as  
datasource from ',pipeline().parameters.tablename)
```



Additionally, we will update the lookup query to insert logs into the audit table for Full load as shown below.

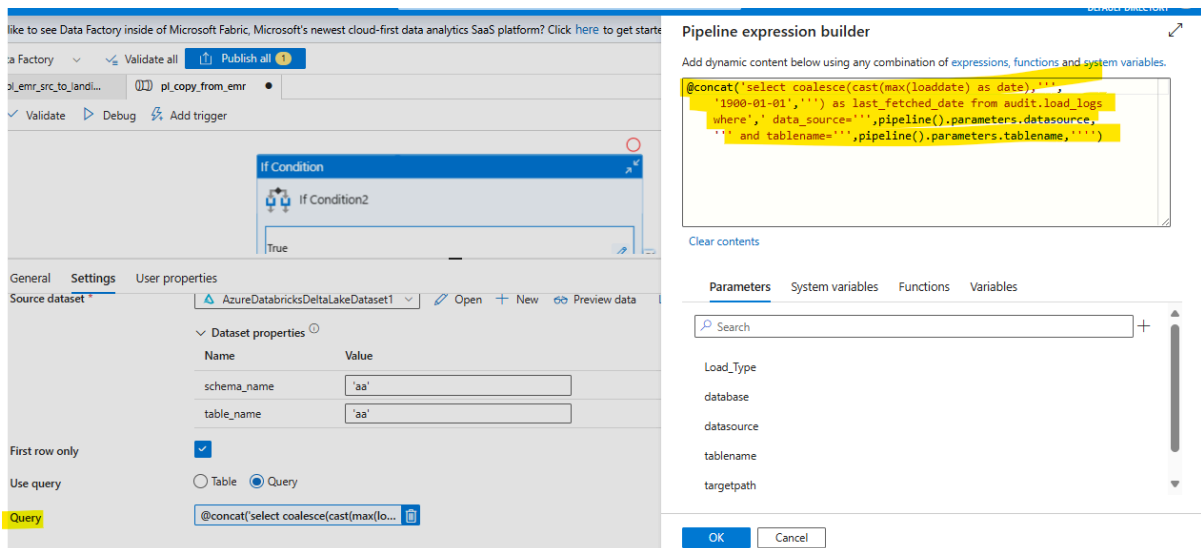
```
@concat('insert into
audit.load_logs(data_source,tablename,numberofrowscopied,watermarkcolumn
nname,loaddate) values (''',pipeline().parameters.datasource, '',
'',pipeline().parameters.tablename, '','',activity('Incremental_Load_
CP').output.rowscopied, '','',pipeline().parameters.watermark, '','',
utcNow(), '''))')
```



Incremental load:

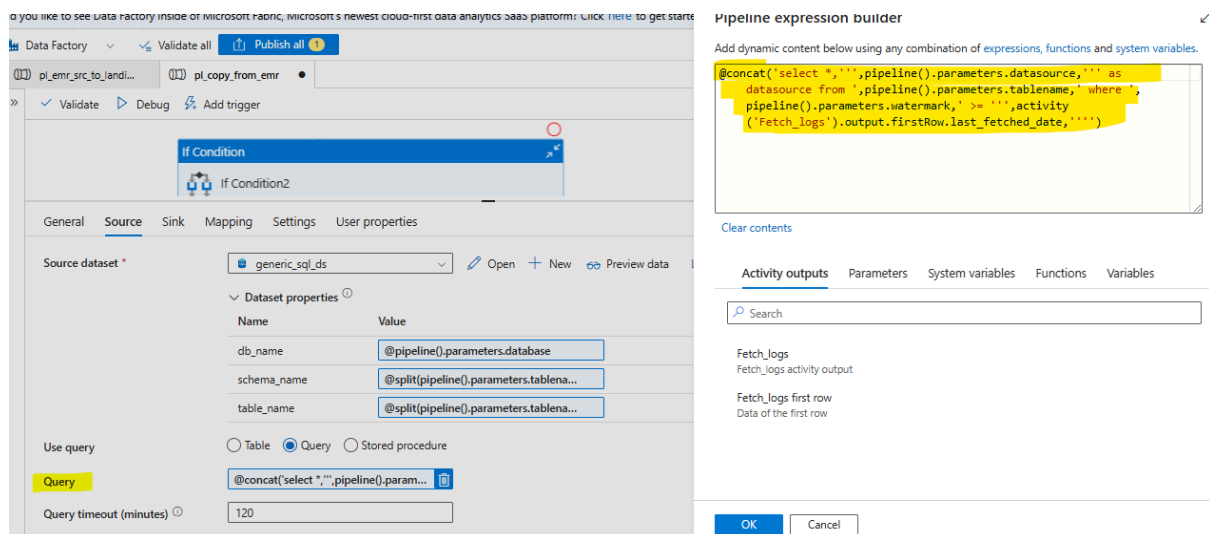
For activity “Fetch logs” we have to update the query instead of item().

```
@concat('select coalesce(cast(max(loaddate) as date),'','1900-01-01','') as last_fetched_date from audit.load_logs where',' data_source=''',pipeline().parameters.datasourcesource,''' and tablename=''',pipeline().parameters.tablename,'''')
```



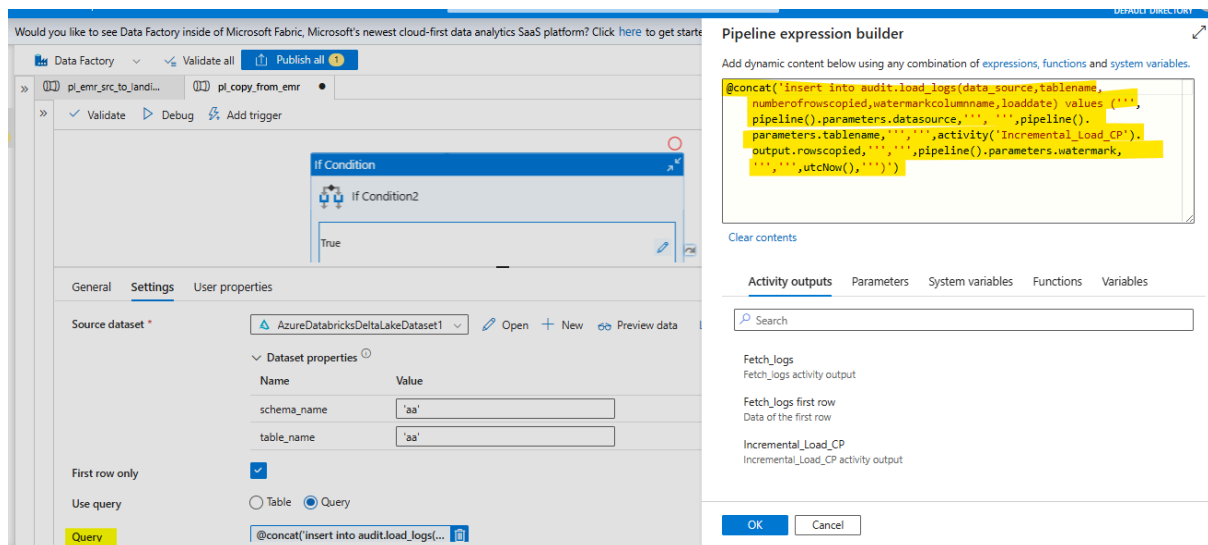
Also we have to update the queries for incremental load as shown below

```
@concat('select *',''',pipeline().parameters.datasourcesource,''' as datasourcesource from ','pipeline().parameters.tablename,' where ','pipeline().parameters.watermark,' >= ','activity('Fetch_logs').output.firstRow.last_fetched_date,'''')
```

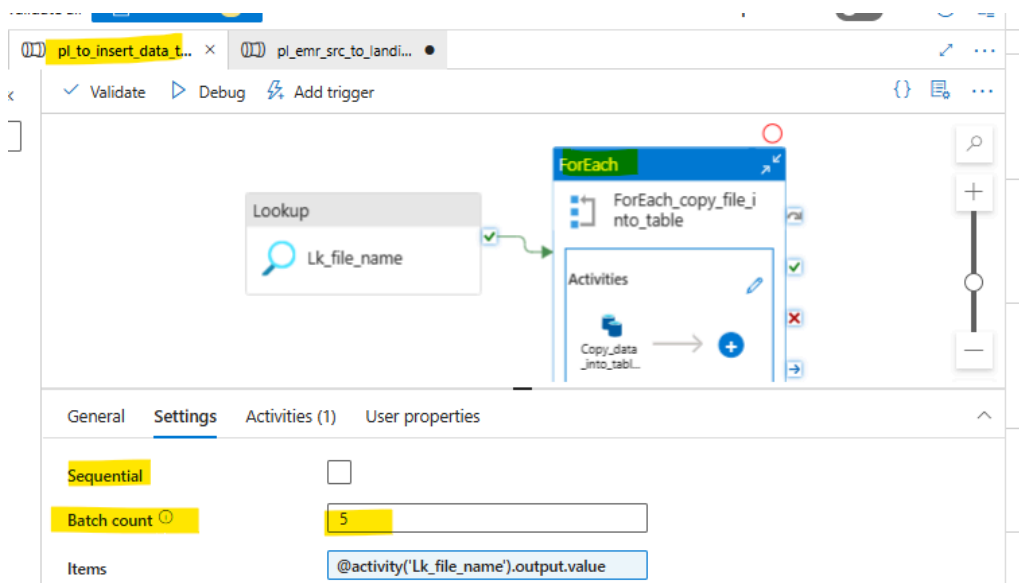


Additionally, we will update the lookup query to insert logs into the audit table for Incremental load as shown below.

```
@concat('insert into
audit.load_logs(data_source,tablename,numberofrowscopied,watermarkcolumn
name,loaddate) values (','pipeline().parameters.datasourcesource',' ',
','pipeline().parameters.tablename',' ','',activity('Incremental_Load_
CP').output.rowscopied,',' ','',pipeline().parameters.watermark,',' ','',
utcNow(),',' '))')
```



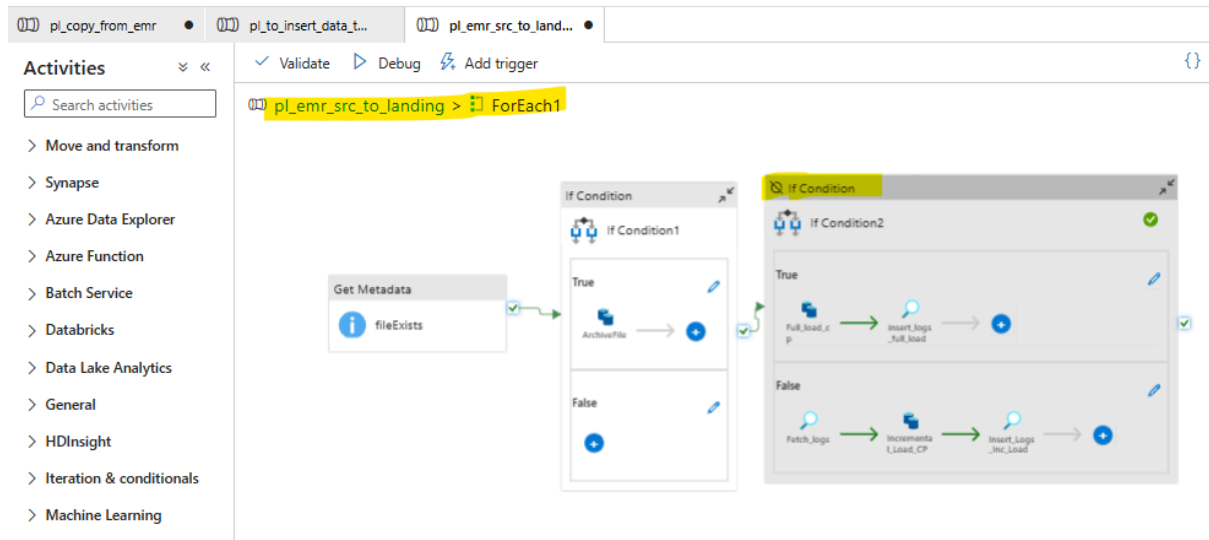
In order to make pipeline -pl_emr_src_to_landing parallel, for “For each” Activity in the pipeline “pl_emr_src_to_landing” we will remove sequential options and will add batch count as 5.



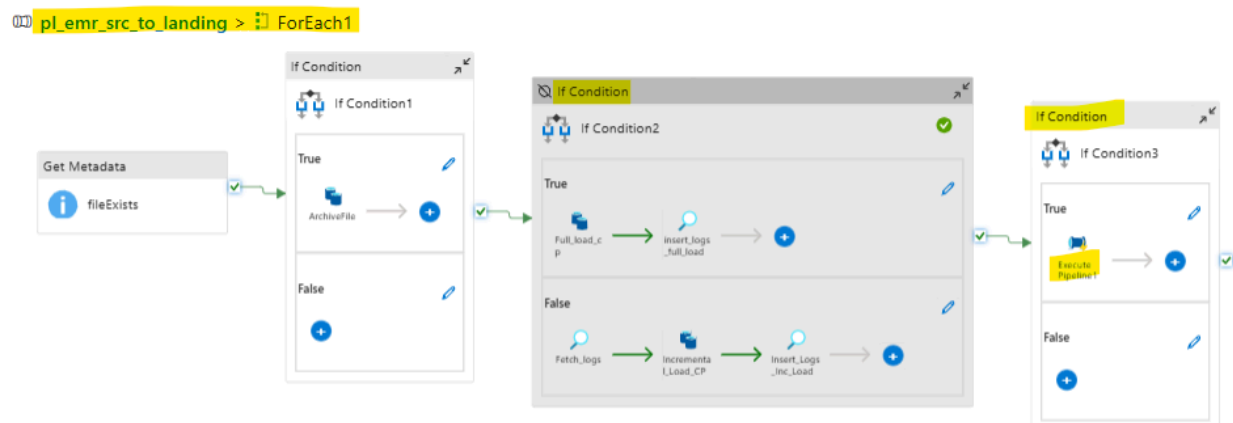
Once this pipeline is updated, publish these changes.

2. Updating the pipeline - pl_emr_src_to_landing

In this pipeline “pl_emr_src_to_landing” deactivate this activity as shown below



Now in this pipeline “pl_emr_src_to_landing” we will add one more if condition with logic `@equals(item().is_active,'1')` as shown below and in this we will add the execute pipeline in which we will attach the recently created pipeline “pl_copy_from_emr” as shown below



Also for this execute pipeline we will pass the parameter value as shown below.

pl_copy_from_emr

Validate Debug Add trigger

pl_emr_src_to_landing > ForEach1 > If Condition3 - True activities

Execute Pipeline

Execute Pipeline1
pl_copy_from_emr

General Settings User properties

Invoked pipeline * pl_copy_from_emr Open + New

Wait on completion ☒

Parameters

Name	Type	Value	Default value
Load_Type	String	@item().loadtype	
database	string	@item().database	
tablename	string	@item().tablename	
datasource	string	@item().datasource	
targetpath	string	@item().targetpath	
watermark	string	@item().watermark	

Steps to move NPI and ICD from API to bronze layer:

For this, we will use the provided code notebook and implement the logic in it.

2. API extracts

=> ICD Code API extract

=> NPI API extract

Note: Run this code notebooks in order to fetch data before going ahead.

Steps to move Claims and CPT data from landing to bronze layer:

As part of the background activity, we will manually place the claims and CPT data files into their respective folders within the landing folder.

Claims data:

Home > Storage accounts > ttadlsdevnew | Containers >

landing ...
Container

Search

Upload Add Directory Refresh Rename Delete Change tier Acquire I

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing / claims

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier	Arch
☐ [.]			
☐ hospital1_claim_data.csv	12/17/2024, 11:00:56...	Hot (Inferred)	
☐ hospital2_claim_data.csv	12/17/2024, 11:00:56...	Hot (Inferred)	

CPT codes data:

Home > Storage accounts > ttadlsdevnew | Containers >

landing ...
Container

Search

Upload Add Directory Refresh Rename Delete Change tier Ac

Overview

Diagnose and solve problems

Access Control (IAM)

Settings

Authentication method: Access key (Switch to Microsoft Entra user account)

Location: landing / cptcodes

Search blobs by prefix (case-sensitive)

Name	Modified	Access tier
☐ [.]		
☐ cptcodes.csv	12/17/2024, 11:02:19...	Hot (Inferred)

Now using the logic present in the code notebooks we will move this data to the bronze layer in parquet format.

3. Silver :

=> Claims

=> CPT codes

Moving data from Bronze layer to Silver.

In this layer, we implement the following logic to transform and refine the data:

Data Cleaning (clean):

=> Removed invalid, null, or duplicate records to ensure data quality.

=> Standardized the data format to align with the Common Data Model for consistency and compatibility across systems.

=> Implement SCD Type 2 logic to maintain historical changes in the data, enabling tracking of changes over time.

Delta Table:

=> Stored the transformed data in Delta tables to support ACID transactions, incremental loads, and versioned data.

This transformation ensures that data from the Bronze layer is cleansed, standardized, and enriched before moving to the Silver layer for further analytics or downstream processing.

We have documented the logic for this transformation in the notebook located in the Silver folder within our Databricks workspace. We have shared the code notebook on Github to which you can refer.

Note: For both claims and CPT codes, we have used the same notebook. This notebook contains the code for moving data from the landing layer to the bronze layer, followed by the necessary transformations to clean and move the data from bronze to the silver layer.

Path: Notebooks → Silver → claims, CPT codes

Moving data from Silver to Gold layer:

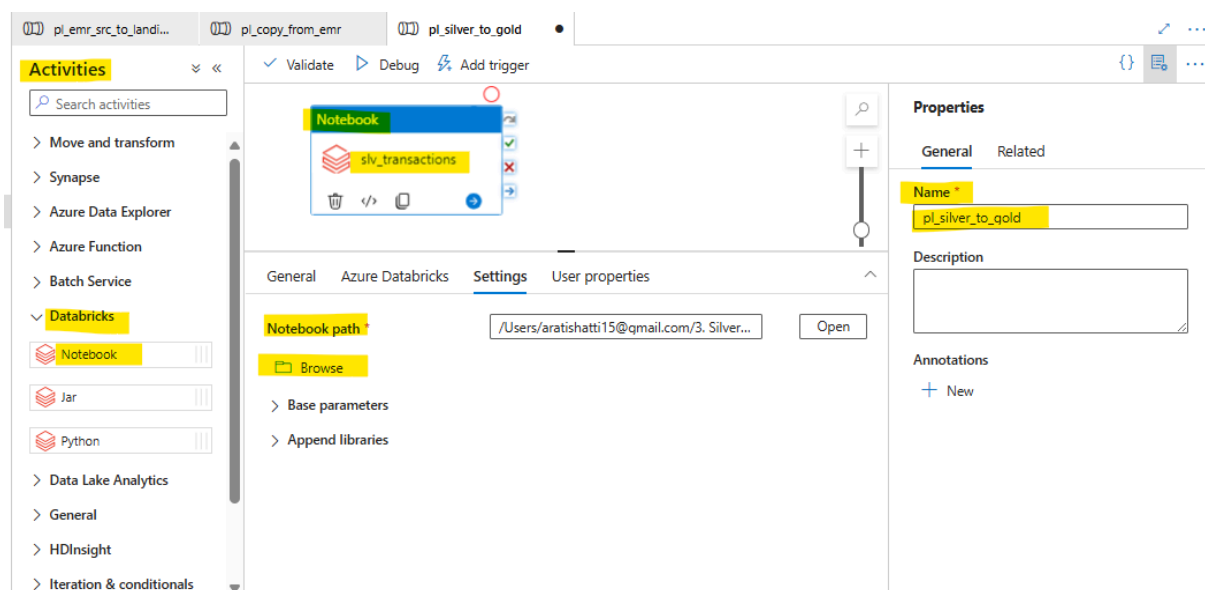
We have created notebooks to implement the required logic, and all these notebooks are located in the "Gold" folder. We have shared the code notebook on Github to which you can refer.

Pipeline to Move data from Silver to Gold layer:

We will create a pipeline “pl_silver_to_gold” to move the data from the Silver layer to the Gold layer.

First create the pipeline “pl_silver_to_gold”, and add the activity Databricks notebook as shown below.

Ex: Here, we have added a notebook activity and named it slv_transaction. Using the Browse Path option, we selected the Transactions notebook located in the Silver folder. Refer below screenshot for more information.



Similarly, we will add the remaining notebooks as demonstrated by Sumit Sir in the video. And our complete pipeline will be as shown below.

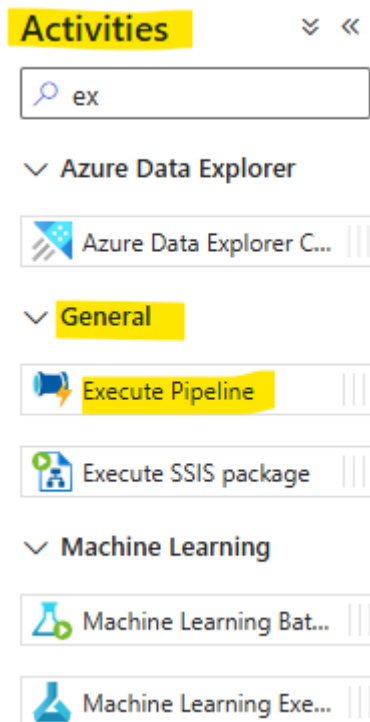


Once this pipeline is created, publish these changes.

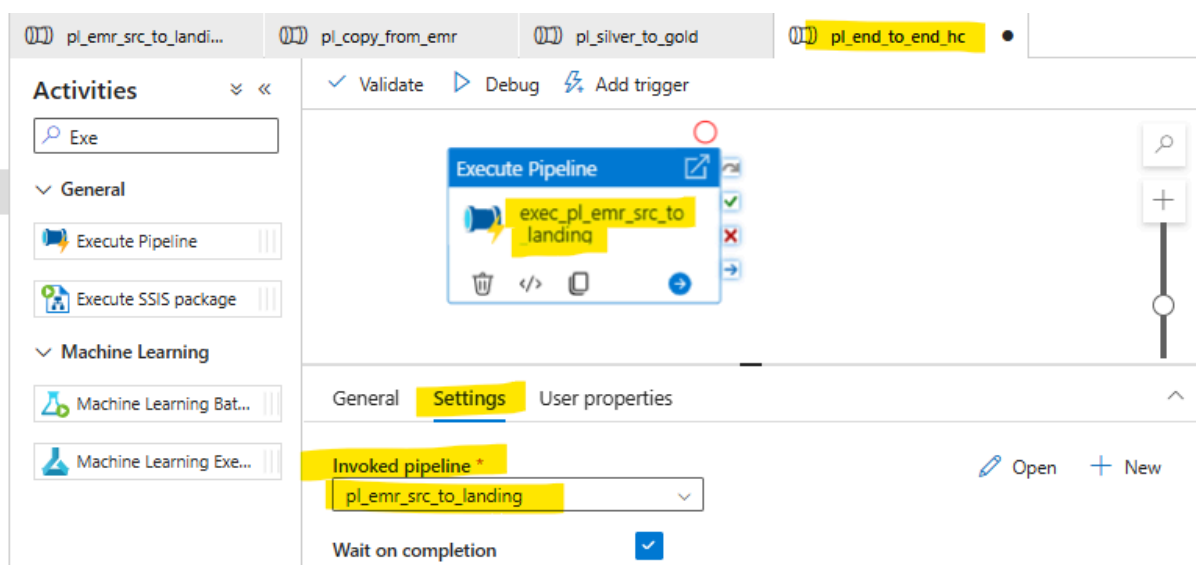
Final Pipeline:

The final pipeline (pl_end_to_end_hc) will include two execution pipelines:

In this we will use an activity Execution pipeline.



=> The first execution pipeline (exec_pl_emr_src_to_landing) will move data from the SQL database to the landing folder.



=> The second execution pipeline (exec_pl_silver_to_gold) will move the cleaned and transformed data from the Silver layer to the Gold layer.

The screenshot shows the Azure Data Studio interface with the 'Execute Pipeline' activity configured. The 'Settings' tab is selected, showing the 'Invoked pipeline' dropdown set to 'pl_silver_to_gold' and the 'Wait on completion' checkbox checked. The 'Activities' pane on the left shows the 'Execute Pipeline' activity selected.

Before running these pipelines make sure to create the silver and gold schema using below commands in catalog `tt_hc_adb_ws`.

```
CREATE SCHEMA IF NOT EXISTS tt_hc_adb_ws.silver;  
  
CREATE SCHEMA IF NOT EXISTS tt_hc_adb_ws.gold;
```

Upon successful execution of the pipeline, the results will be displayed as shown below.

The screenshot shows the 'Activity runs' section for the 'pl_end_to_end_hc' pipeline. The 'List' tab is active, showing a table of activity runs. The table has columns for Activity name, Activity status, Activity type, Run start, Duration, Integration runtime, User properties, and Activity run ID. Two activity runs are listed, both with a status of 'Succeeded'.

Activity name	Activity status	Activity type	Run start	Duration	Integration runtime	User properties	Activity run ID
exec_pl_silver_to_gold	Succeeded	Execute Pipeline	12/18/2024, 9:49:00 AM	3m 2s			a2d7f94f-7d47-4e25-9f...
exec_pl_emr_src_to_landing	Succeeded	Execute Pipeline	12/18/2024, 9:30:17 AM	18m 44s			b0607903-b0df-4742-a...

To link a project on Github follow below steps.

In Azure Databricks: Get the username and token for your databricks

In Github:

Get the username of your github account.

Generate personal access token:

Go to Github => Settings => Developer Settings => Personal Access token => Tokens (Classic) => select all the scopes => generate token => Copy the token

You can provide below permissions:

repo – Full access to repositories.

read:org – Access organizational-level data.

workflow – Trigger GitHub Actions workflows.

read:packages (Optional, for package access).

New personal access token (classic)

Personal access tokens (classic) function like ordinary OAuth access tokens. They can be used instead of : over HTTPS, or can be used to [authenticate to the API over Basic Authentication](#).

Note

github

What's this token for?

Expiration *

30 days ▾ The token will expire on Fri, Jan 17 2025

Select scopes

Scopes define the access for personal tokens. [Read more about OAuth scopes](#).

☒ repo	Full control of private repositories
☒ repo:status	Access commit status
☒ repo_deployment	Access deployment status
☒ public_repo	Access public repositories
☒ repo:invite	Access repository invitations
☒ security_events	Read and write security events

Create the token and note it once created.

Now go to databricks => setting => Linked accounts = > Git provider (Github) then select personal access token => then provide the username and the token for github

Settings

- Workspace admin
- Appearance
- Identity and access
- Security
- Compute
- Development
- Notifications
- Advanced
- User
 - Profile
 - Preferences
 - Developer
 - Linked accounts
 - Notifications

Linked accounts

Connect your Databricks account to other services

Git integration

With co-versioned repo

Databricks Git folders and Repos allow you to clone a remote Git repo, which you can specify when you add a Git folder. [Learn more](#)

With individual notebooks

Although we recommended using co-versioned repo for Git integration, Databricks supports individual notebook version control integration with GitHub, Bitbucket Cloud, or Azure DevOps Services (using AAD authentication only).

Set your Git provider and credentials

You can also set your Git provider credentials via API. [Learn more](#)

Git provider

GitHub

☐ Link Git account
Simple setup in a few clicks to link your GitHub account. [Learn more](#)

☒ Personal access token
Use an existing personal access token to authenticate Git requests.

Git provider username or email

.....

Token

To generate a GitHub personal access token, follow the [GitHub documentation](#). The token must have the "repo" and "workflow" (if your repository has GitHub Action workflows) scopes.

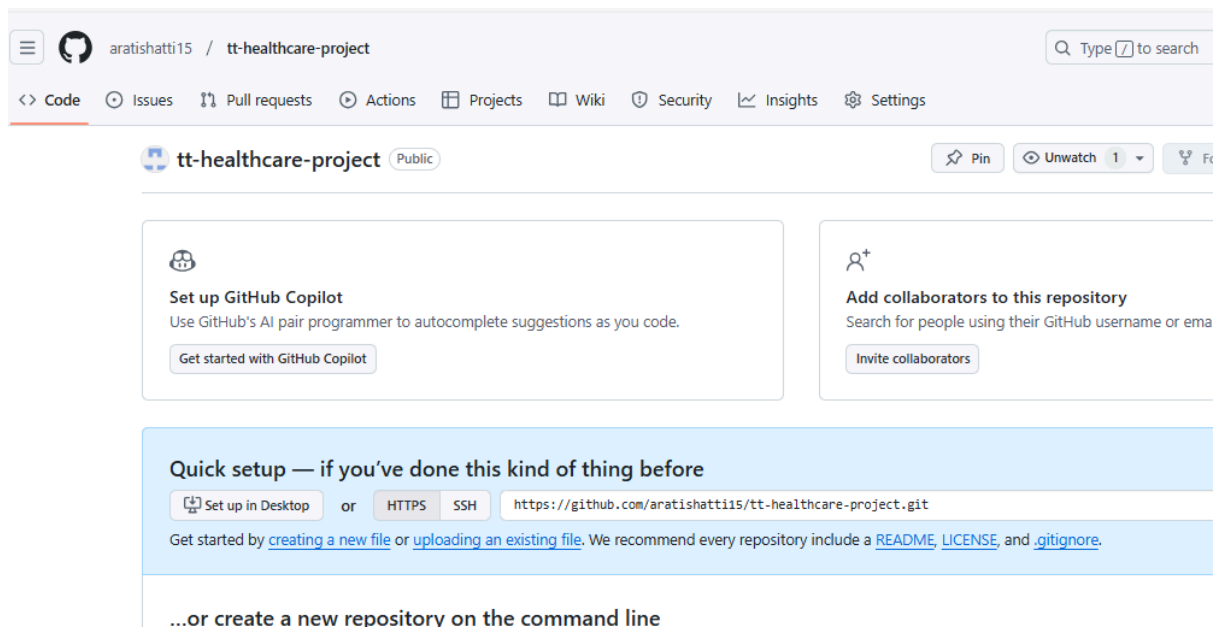
.....

Save

Now create the repository for healthcare project on github account:

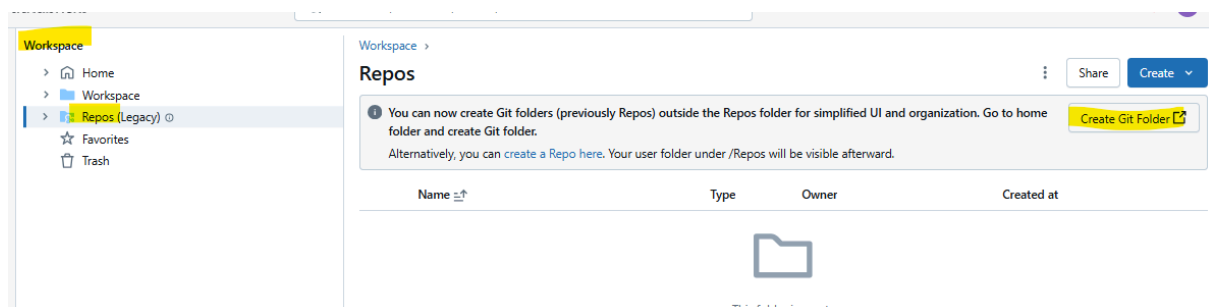
To create a repository on GitHub:

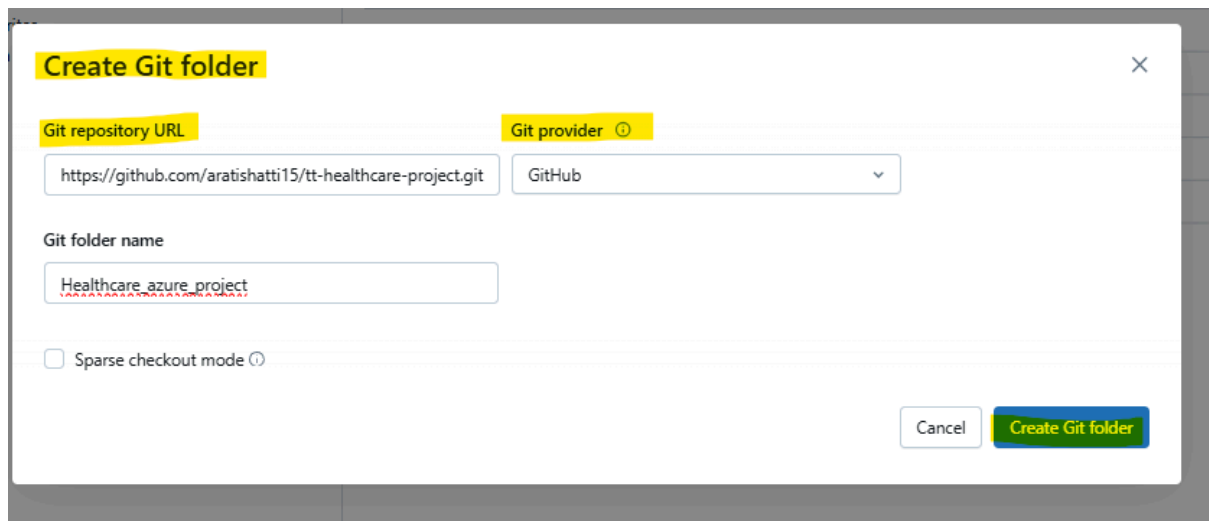
1. Go to [GitHub](#).
2. Click on **New** under "Repositories" in your GitHub profile.
3. Provide:
 - Repository Name: e.g., **tt-healthcare-project**.
 - Description (optional).
 - Choose **Public** or **Private** visibility.
4. Click **Create repository**.



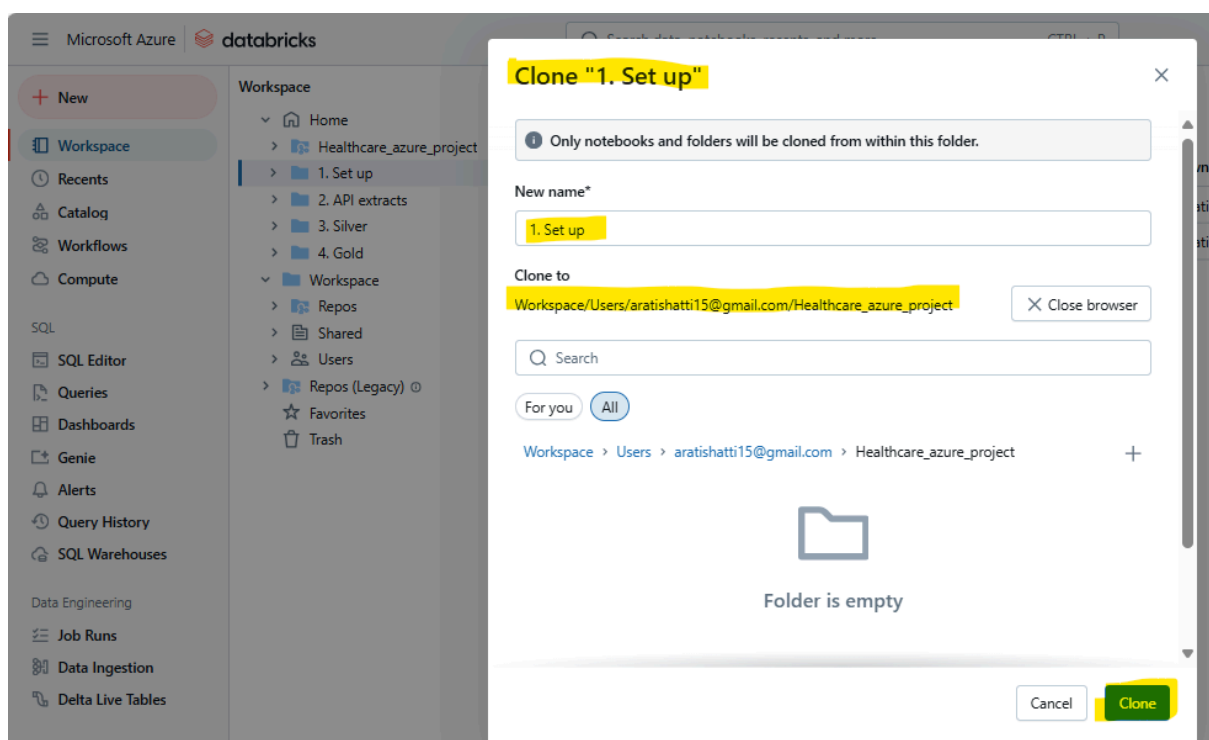
Cloning GitHub Repository into Databricks

Go to databricks => Workspace => Repos => select option create git folder





Now you can clone existing folders and files into this git folder.
Right click on file/folder => clone => then using the browse option select this git folder.



Similarly clone all the required files and folders.

To use Git in Databricks, click on the three dots next to the Git-linked folder and select the Git option. This allows you to pull and push changes, as well as create branches, directly from the Databricks interface.

Healthcare_azure_project

main ☆



Share

Create ▾

Name ↕	Type	Owner		
1. Set up	Folder	Arati Hatti		4, 01:56:26 AM
2. API extracts	Folder	Arati Hatti		4, 09:00:02 AM
3. Silver	Folder	Arati Hatti		4, 09:00:23 AM
4. Gold	Folder	Arati Hatti		4, 09:00:39 AM

Git...

Import

Download as

Copy URL/path

Rename

Move

Add to favorites

Move to Trash